

The Morphosyntactic Pivot Hypothesis: Hierarchical Parametric Organization of the Locus of the EPP

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1. Introduction

1.1 The problem

Many theories presuppose that T universally hosts the EPP (Chomsky 1995, 2000). This assumption has already been challenged from within Minimalism itself. Alexiadou & Anagnostopoulou (1998) show that languages with sufficiently rich verbal agreement can satisfy the EPP through head-movement of V to Agr rather than through movement of an XP to Spec-TP, locating EPP-satisfaction on Agr rather than T in pro-drop, VSO/VOS languages such as Greek, Arabic, and the Celtic languages. Bobaljik & Jonas (1996) and Bobaljik & Thráinsson (1998) independently argue that the availability of an additional subject-licensing position above VP is itself parametrized by the richness of verbal agreement morphology — the Rich Agreement Hypothesis — linking functional structure directly to morphological paradigm richness rather than treating T as a fixed, universal locus. Miyagawa (2010) extends this line further, proposing that languages fall into two macro-types depending on whether the EPP is discharged through ϕ -feature agreement or through discourse-configurational (topic/focus) marking on C (§4.3 returns to this proposal directly).

What remains unresolved across this literature is not whether the EPP-host varies — that variation is already established — but how to identify the pivot in a given language without circularity, and how far the variation extends beyond the T-versus-Agr and Agreement-versus-Discourse contrasts these proposals are built on. This paper develops a non-circular, cross-diagnostic identification protocol — convergent syntactic diagnostics corroborated by independent morphological criteria — and applies it to a typologically broader sample in which the pivot proves to be hosted not only by T or Agr but also by Mood, Aspect, and Aspect/Evidentiality. The resulting five-way typology is then shown to reduce to a single hierarchical two-parameter architecture (§3), rather than remaining a language-by-language parametric list.

1.2 Overview of the Study

This paper develops the Biphasic Identification Protocol (§1.4), applies it systematically to identify morphosyntactic pivots in six core languages (§2), discovers that variation exhibits Hierarchical Parametric Organization (§3), validates this with 82-language statistical analysis achieving 96.60% PC1 (§3.5), and explores theoretical implications (§4-5). The key finding is that the EPP locus exhibits hierarchical rather than flat parametric structure, as revealed through independent diagnostic methodology.

1.3 The Morphosyntactic Pivot Hypothesis (MPH)

The EPP bundle is not universally tied to T but to the morphosyntactic pivot—the functional head that (i) independently meets syntactic diagnostics of structural

¹ **Supplementary Materials and Reproducibility Package**

The complete Supplementary Materials, datasets, analysis scripts, elicitation materials, and replication resources are permanently archived in the Open Science Framework (OSF):

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centrality and (ii) is corroborated by morphological prominence². That pivot projects obligatorily, values the goal's [uCase] and hosts [uEPP].

The morphosyntactic pivot is identified independently of the categories it realizes. In φ -available systems, it hosts the productive φ -feature system; in φ -unavailable systems, it may instead realize non- φ categories such as aspect, evidentiality, mood, or tense. The identity of the pivot is therefore logically distinct from its feature inventory.

1.4 Feature System and Theoretical Background

The analysis assumes standard Minimalist feature theory (Chomsky 2000, 2001, 2007; Pesetsky & Torrego 2007). Functional heads bear features that drive syntactic operations and must satisfy interface conditions. We adopt the following feature typology:

Feature interpretability:

- **[iF] (interpretable features)**: Semantically contentful; persist to LF and PF
- **[uF] (uninterpretable features)**: Formal; must be valued/deleted before transfer to interfaces
- **φ -Availability**: the availability of semantic content at LF. This property, together with three further diagnostics introduced in §3.5, constitutes empirical evidence for a language's φ -state (defined formally in §3.1).

Relevant features:

- **[uEPP]**: Uninterpretable EPP feature requiring satisfaction via movement to the head's specifier
- **[iCase]**: Interpretable case feature on functional heads (assigns case to DPs)
- **[uCase]**: Uninterpretable Case feature on DPs (requires valuation)
- **[i φ]**: Interpretable φ -features (person, number, gender/noun class); persist to interfaces
- **[u φ]**: Uninterpretable φ -features; require valuation and deletion
- **[n φ]**: Absence of φ -features on the pivot

Key principle: Interpretable features persist through the derivation to both LF (semantic interpretation) and PF (morphological realization). [u φ], being 'defects' must be eliminated (via valuation and deletion) before transfer to avoid interface crashes (Full Interpretation; Chomsky 1995).

φ -features and case-based intervention: The presence of φ -features on a functional head does not guarantee surface agreement morphology. In ergative systems, ergative case on the external argument blocks φ -Agree between the pivot and the subject DP (Bobaljik 1993, Laka 1993, Woolford 1997, Anand & Nevins 2006). When the pivot bears [i φ] but the goal bears ergative case, the pivot's [i φ] cannot establish matching relations with the goal's [i φ]. However, the subject DP's φ -features remain recoverable despite no matching with the pivot. If a nominative internal argument is accessible, the pivot may agree with it instead, yielding object agreement.

Relevance to pivot identification:

The pivot head hosts [iCase] and [uEPP] but varies cross-linguistically in its φ -specification. This variation correlates with morphological patterns and subject-licensing properties:

- Pivots with [u φ] (e.g., English T): impoverished agreement, no pro-drop
- Pivots with [i φ] (e.g., Swahili Agr, Hindi-Urdu Asp): rich agreement, pro-drop possible

² I apply Bybee's (1985) diagnostics, i.e., (obligatoriness, relevance, paradigmatic richness, generality) to determine morphological prominence for every language in the sample.

- Pivots with [φ -less] (e.g., Balti Asp/Evid): no agreement, discourse/topic based pro-drop

This featural variation, its formal representation, and its interface consequences will be developed in detail in Section 3.

1.5 A Biphasic Diagnostic Protocol for Pivot Identification

Moving from the theoretical construct of the morphosyntactic pivot to its empirical identification requires a method that insulates structural diagnosis from morphological preconceptions. I therefore propose a Biphasic Diagnostic Protocol, which establishes pivothood as the point of convergence between a head's clause-delimiting syntactic properties and its featural prominence in the morphological paradigm.

Both the phases are essentially independent of each other. A functional head must pass both independent phases to be formally designated as the morphosyntactic pivot.

The independence of Phase 1 and Phase 2 is demonstrated by languages like Czech and Finnish, which show complete syntactic clustering (Rich_Agreement = 1, Morph_ProDrop = 1, No_Expletive = 1) identical to Type 3 languages like Italian and Spanish and yet receive Phi_Availability = 0.2 rather than 1.0. This distinction reflects Phase 1's independent assessment of whether φ -features on the morphosyntactic pivot are interpretable [$i\varphi$] (carrying semantic content to LF, as in Italian/Spanish) or uninterpretable [$u\varphi$] (formal requirements deleted before LF, as in Czech/Finnish). The fact that languages with identical Phase 2 profiles receive different Phase 1 classifications demonstrates that the two phases assess distinct empirical domains, making circular coding impossible.

1.5.1 The Structural Autonomy Diagnostics

- **Clause-boundary diagnostics**
 - *Coordination (ConjP)*: which projection can form independent conjuncts?
 - *Complementizer selection*: which projection is selected by complementizers?
 - *Ellipsis/fragment diagnostics*: which projection survives in fragmentary answers?
- **Movement diagnostics**
 - *Subject landing site*: adverb ordering, binding, and surface subject position.
 - *Extraction/intervention*: which projections intervene in wh-movement / topicalization?
 - *Freezing effects*: whether extraction out of a moved subject is degraded.

Nomination Rule (operational). A projection is nominated as the *syntactic candidate pivot* when it clearly passes at least two clause-boundary diagnostics *and* at least two movement diagnostics. This binary, replicable rule reduces subjectivity and makes identification falsifiable (see Appendix A.2)

1.5.2 The Morphological Prominence Diagnostics

Once syntax nominates a candidate, I apply morphological diagnostics (Bybee 1985) as independent corroboration:

- **Obligatoriness**: the marker projects in most finite clauses (operational threshold: $\geq 90\%$ of attested finite clauses in naturalistic data or unanimous native-judgment).
- **Relevance (functional tie)**: the morphology is functionally tied to clause-level interpretations (tense/aspect/mood/evidentiality/agreement) and argument licensing.

- **Paradigmatic richness:** the morphosyntactic category displays multiple contrastive values (operational threshold: at least three distinct values relevant to clause-typing or ϕ -expression).
- **Generality/Productivity:** the morphology occurs across verb classes and clause types (operational threshold: productive for $\geq 80\%$ of attested verb types in a sample).

If morphology corroborates the syntactic candidate (strong match on two or more Bybee criteria), we accept that head as the morphosyntactic pivot. If there is divergence, the mismatch is reported and analysed (e.g., erosion, fusion, mixed pivot systems). Importantly, the morphosyntactic pivot is identified before parameter assignment. The subsequent parameters characterize the already identified pivot rather than determining its identity (see Appendix A.3).

1.6 Research Methodology

This study integrates diagnostic protocols with quantitative validation to test the Morphosyntactic Pivot Hypothesis on six core languages from different families/subfamilies. To demonstrate the morphosyntactic consequence of the MPH, i.e., Hierarchical Parametric Organization in the locus of the EPP host, the sample is extended to fifty-two genealogically unrelated languages including the six core languages.

1.6.1 Sample Selection

The core sample comprises six languages from distinct families/sub-families, i.e., English, Hindi-Urdu, Swahili, Italian, Balti, and Warlpiri, representing five continents (Hindi and Urdu, though drawn from a shared speaker sample, are coded as two separate dataset entries per standard typological practice, yielding $N=7$ in the quantitative analyses of §3.5 and Appendix D). Five languages were investigated through primary data collection ($N=110$ speakers total); Warlpiri analysis synthesizes published descriptions. Systematic expansion with forty-six additional languages selected for genealogical diversity and typological coverage, tests the hierarchical parametric structure. This design balances genealogical diversity with analytical tractability (see Appendix B & C).

1.6.2 Theoretical Integration

The empirical results (§2) inform the theoretical development (§3), where I propose a Hierarchical Parametric Organization which consists of two binary parameters in a logical-dependency relation:

Level 1: ϕ -Availability Parameter [$\pm\phi$]

Does the pivot bear ϕ -features?

- [$-\phi$] $\rightarrow$ Type 1 (ϕ -less): No ϕ -features (e.g., Balti) $\rightarrow$ Type 1 ('0.1' at Phi_Availability)

- [$+\phi$] $\rightarrow$ Proceed to Level 2

Level 2: ϕ -Interpretability Parameter [$\pm\text{interp}$] (only if [$+\phi$])

Are ϕ -features interpretable at LF?

- [$-\text{interp}$] $\rightarrow$ Type 2 ([$u\phi$]): Uninterpretable ϕ (e.g., English) $\rightarrow$ 0.2 Phi_Availability

- [$+\text{interp}$] $\rightarrow$ Type 3 ([$i\phi$]): Interpretable ϕ (e.g., Hindi-Urdu, Swahili, Italian, Warlpiri) $\rightarrow$ 1.0 Phi_Availability

By logical-dependency, I refer to a relation where the value of one parameter determines the value of the lower parameter. This hierarchical dependency reduces the logical space from $2^2 = 4$ combinations to 3 attested types, as [$-\phi, \pm\text{interp}$] is incoherent (absent features cannot be interpretable or uninterpretable). This is an

ontological/computational asymmetry, not a stipulated convention — existence necessarily precedes computation over what exists (§3.3).

1.6.3 Statistical Validation of Parametric Dependencies

To test whether this hierarchical architecture correctly captures the organization of morphosyntactic variance, I employ Principal Component Analysis (§3.5) on four independently documented phenomena³: (i) ϕ -Availability, (ii) agreement morphology richness, (iii) morphological pro-drop licensing, (iv) and expletive absence. These variables operationalize the properties predicted to cluster under the ϕ -Availability Parameter and the ϕ -Interpretability Parameter.

To avoid any possibility of circularity, it is crucial that the four variables remain operationally independent. Each variable targets a distinct empirical domain: *Phi_Availability* is diagnosed via bound-variable licensing and coordinated-subject agreement resolution (Appendix F §2.2), neither of which references null-argument licensing, agreement richness, or expletive status. *Rich_Agreement* measures morphological exponence of person/number distinctions independent of their interpretive status; *Morph_ProDrop* and *No_Expletive* are diagnosed via separate elicitation batteries (Appendix D) targeting null-subject licensing and expletive-construction acceptability respectively. This operational independence may diverge empirically (e.g., Type 2b languages with ϕ -availability but impoverished morphology, or TAM-rich systems with rich paradigms but no ϕ -features; see Appendix B.2), and ensures that any observed correlation in §3.5 reflects genuine convergence on a shared latent ϕ -state (§3.1) rather than shared operational basis.

Theoretical Predictions:

If the Hierarchical Parametric Organization is correct, the statistical analysis should reveal:

P1: Dominant dimensional structure (PC1 > 70%). A language's ϕ -state should constitute the primary organizing dimension. This tests whether the properties traditionally treated as independent (agreement, pro-drop, expletives), in fact derive from a single parametric choice.

P2: High internal consistency ($\alpha > 0.80$). The four diagnostic variables should form a coherent syndrome, confirming they measure a unified construct rather than orthogonal phenomena.

P3: Non-monotonic behavioural clustering tracking ϕ -Interpretability, not mere ϕ -presence. Type 1 (ϕ -less) and Type 3 (interpretable ϕ) both permit null subjects and disallow expletives — Type 1 via discourse-pragmatic licensing in the total absence of a ϕ -probe, Type 3 via morphological licensing through an interpretable ϕ -probe. Type 2 (ϕ -present but uninterpretable) is the sole type that blocks null-subject licensing and requires expletives: it possesses a ϕ -probe, unlike Type 1, but that probe delivers no interpretable content capable of licensing recovery of a null subject, unlike Type 3. The predicted pattern is therefore U-shaped in ϕ -Availability rather than monotonic: Type 2 is not an intermediate point between Type 1 and Type 3, but the qualitatively most restrictive type of the three, precisely because it combines the syntactic cost of a ϕ -probe with none of its interpretive benefit — the asymmetry developed in §3.3.

³ These phenomena have independent research histories: Corbett (2006) on agreement, Rizzi (1982) on pro-drop, Chomsky (1981) on expletives, and Chomsky (1995) on ϕ -features. Their high correlation ($r = 0.78\text{--}1.00$) is an empirical discovery demonstrating convergent validity.

P4: A flat, non-hierarchical alternative in which ϕ -Availability alone determined pro-drop and expletive behavior would instead predict a monotonic gradient (Type 1 $\leq$ Type 2 $\leq$ Type 3), with Type 2 patterning as an intermediate case rather than the outlier. Finding the U-shaped pattern instead of the monotonic one is therefore a distinguishing empirical test of the hierarchical model, confirmed by the type profiles in §2.7 and Table 4 (§3.5).

Crucially, the statistical validation tests the logical-dependency structure directly. If the two parameters were independent (flat architecture), the four logically possible combinations would be expected to distribute across two comparably weighted dimensions, yielding multidimensional structure (PC1 $\approx$ 35–50%). If instead ϕ -Interpretability is strictly dependent on ϕ -Availability — Level 2 is defined only where Level 1 = [+ ϕ] — the valid state space collapses from four cells to three, and those three states fall on a single ordinal scale (0.1 < 0.2 < 1.0) rather than an unordered set. This predicts unidimensional structure (PC1 > 70%) with three discrete clusters positioned along that scale, not two. The concentration of variance in PC1 (96.60%) and PC2 (3.40%), together with the three empirically distinct types confirmed in §3.5 (Table 4) and the U-shaped pro-drop/expletive pattern above, confirms this, validating the hierarchical dependency empirically.

1.6.4 Independent Corpus Validation via Universal Dependencies

To further guard against circularity, the hand-coded Phi_Availability values were tested against an entirely independent source: raw morphological data extracted from the Universal Dependencies treebank corpus (v2.17, 339 treebanks, 186 languages), a resource built by hundreds of annotators across two decades with no relationship to MPH. A corpus-derived ϕ -index — a mechanical, zero-learned-parameter count of Person/Number/Gender feature co-occurrence on finite verbs — was computed for 80 of the 82 sample languages (excluded: Swahili, Balti, validated instead via primary elicitation data per §1.6.1) and compared against the gold Phi_Availability coding using a fixed, theory-derived threshold set a priori, never fit to the gold labels. Because the ϕ -index measures only morphological richness while MPH's diagnostics for interpretability are syntactic (pro-drop licensing, expletive obligatoriness), the theory's own architecture predicts, in advance of running this comparison, exactly two failure modes for the proxy and no others. First, languages whose ϕ -features remain syntactically available, but whose agreement morphology has eroded should appear as false negatives at the availability level — the proxy will read them as phi-less despite phi being syntactically active. Second, languages whose ϕ -features have become uninterpretable while their agreement morphology survives diachronically (the Type 2b profile) should appear as false positives at the interpretability level — the proxy will read them as interpretable because it cannot see that morphology has decoupled from interpretability. No other failure pattern is predicted: the proxy should not misclassify languages at random, nor fail on languages where morphology and the syntactic diagnostics agree.

Results: the ϕ -index recovers ϕ -availability (0.1 vs. {0.2, 1.0}) at 97.5% accuracy (AUC=0.998, N=80) and ϕ -interpretability (0.2 vs. 1.0) at 95.0% accuracy (AUC=0.966, N=80). Critically, the four residual misclassifications fall exactly into the two failure modes predicted above, and no others — not one undifferentiated group of noise, but a structured match to a prediction stated before the comparison was run. Norwegian and Swedish are false negatives at the ϕ -availability level (predicted 0.1/phi-less, gold says 0.2): both have completely lost verbal person-marking morphology, identical in surface

appearance to genuinely phi-less languages like Chinese — yet the syntactic diagnostics reveal phi is nonetheless present, evidenced by the same two hallmarks that identify German, English, and Dutch as 0.2 languages: the absence of morphological pro-drop and the obligatoriness of expletives. Icelandic, German, and Finnish are false positives at the interpretability level (predicted 1.0/interpretable, gold says 0.2) and represent the opposite profile: unlike Norwegian and Swedish, these languages retain genuinely rich verbal agreement (Rich_Agreement=1) — yet richness of morphology does not entail interpretability, precisely the central empirical claim of Type 2b. That the φ -index, built purely from morphological richness, cannot distinguish either case from Type 3 is expected: it has no access to the pro-drop and expletive diagnostics that correctly identify all five languages as 0.2 regardless of whether their agreement morphology is rich or entirely absent. This constitutes a genuine, independent-data test of MPH's core prediction — one that could have failed, in a specific and pre-stated way, and did not fail outside that pre-stated pattern.

1.6.5 Data Availability, Transparency and Reproducibility

The study is designed for full reproducibility. Elicitation materials, raw acceptability-judgment data, morphological codings, the complete eighty-two-language dataset, the Universal Dependencies validation script and treebank-derived φ -index outputs (§1.6.4), and a single R script reproducing every figure and statistic reported are provided in the online Supplementary Materials. All raw datasets, analysis scripts, and reproducibility materials are permanently and publicly available via the Open Science Framework: <https://doi.org/10.17605/OSF.IO/K95P2>, in accordance with journal policy and ethical requirements for anonymized participant data.

Limitations & pre-registered mitigations

- **Sampling limits:** for smaller-language corpora, the obligatoriness threshold is lowered and impacts discussed; sensitivity analyses are reported.
- **Dialectal variation:** where dialect splits are known (e.g., Balti varieties), dialect-specific responses are reported, or collapsed only after verifying homogeneity.
- **Subjectivity in diagnostics:** operational thresholds and dual-coder checks reduce the threat of subjective labelling.

Future evidence may require refinement of individual diagnostics without necessarily affecting the underlying theoretical architecture.

1.7 Empirical Scope

The Biphasic Identification Protocol is applied to the six core languages, with syntactic diagnostics identifying the pivot and morphological diagnostics providing corroborating evidence. Five core languages underwent extensive syntactic testing with native-speaker judgments (N=110); the sixth was analyzed via published description. The seventy-five outgroup languages were not tested under the core protocol; their morphosyntactic profiles were developed from existing descriptive and theoretical literature, cross-checked against Universal Dependencies treebank data where available (§1.6.4). Each case study (§2) follows the same template: (i) syntactic battery results, (ii) morphological corroboration by Bybee criteria, (iii) verdict on pivothood. Comparative synthesis and theory (§3) follow; full diagnostic detail is in Appendix E.

1.8 Consequences

This methodology — syntax-first, morphology-second, explicit thresholds, transparent coding, reproducible analysis, and independent corpus corroboration — mitigates circularity in pivot identification and supports a principled typology.

The Balti case demonstrates an under-documented third parametric type: pivots without any φ -specification ($n\varphi$). Native-speaker judgments (N=30, Skardu fieldwork)

uniformly report no person/number distinctions on aspectual/evidential morphology; the evidential system distinguishes egophoric, inferential, reportative, and visible-evidence categories, but these encode epistemic, not agreement, distinctions. This illustrates a system where case licensing and EPP satisfaction occur independently of φ -Agree (§2.5, Appendix E.5). The absence of productive φ -features does not imply the absence of a morphosyntactic pivot. In φ -unavailable systems, the pivot may instead realize non- φ categories such as evidentiality or aspect.

1.9.1 Theoretical Reformulation of the EPP: The primary contribution is a path toward systematic decoupling of the EPP from Tense and offering a better alternative. I demonstrate that the requirement for a structural anchor is satisfied by the projection that is morphosyntactically most central in a language — the pivot. This explains why languages like Hindi-Urdu exhibit EPP effects (e.g., subject licensing, pro-drop patterns) anchored in Aspect, not Tense, and why Warlpiri satisfies its EPP via head movement of its agreement complex.

1.9.2 A fully operationalized falsifiable minimalist parameter system: I develop a compact set of hierarchically organized φ -related parameters whose values are determined exclusively by independently testable syntactic diagnostics, rather than descriptive generalizations or post hoc typologizing. The Biphasic diagnostic protocol introduced here—Syntax Phase → Morphology Phase—is defined with explicit criteria, threshold conditions, and falsifiable tests. The ordering is not incidental: languages like German and Icelandic, which retain rich verbal agreement morphology yet pattern as [u φ] rather than [i φ], would be misclassified by any protocol that inferred interpretability from morphological richness alone. Because the Syntax Phase determines Phi_Availability independently of the Morphology Phase, the protocol correctly separates these languages from Type 3 despite their superficially rich paradigms — a result that follows from the diagnostic architecture, not from a language-by-language exception list.

This yields maximally objective, reproducible, and coder-independent parameter assignments — but more importantly, it yields a theory that is falsifiable in the classic Popperian sense (Popper 1959): MPH forbids specific, independently checkable observations, and the study's design ensured those observations were genuinely possible outcomes rather than foregone conclusions.

Concretely, MPH predicts that Phi_Availability, Morph_ProDrop, and No_Expletive derive from a single latent φ -state and should therefore converge deterministically — the theory is falsified by any language where these three diverge from one another.

Rich_Agreement is not part of this inviolable core, and its exclusion is principled rather than post hoc: interpretability is established through syntactic diagnostics, not morphological richness, so agreement can persist after interpretability is lost (Type 2b) or erode while phi remains syntactically active (a mirror-image case). This architecture pre-specifies, not merely accommodates, exactly where a morphology-only measure should fail — a prediction tested and confirmed in §1.6.4

1.9.3 A two-level architecture with internal logical dependence: φ -features are organized into two levels — φ -Availability (narrow syntax) and φ -Interpretability at LF (interface visibility) — where Interpretability is defined only if Availability holds. This dependency, not stipulation, is what reduces the logical space from four combinations to three attested macro-types: [- φ , $\pm$ interp] is incoherent, since absent features cannot be interpretable or uninterpretable. The result is a derivationally constrained, minimal typology of φ -systems.

1.9.4 Identification of an under-recognized configuration in existing typological treatments: The model predicts a specific type of language—one with no φ -features in the narrow syntax but with LF-visible nominal interpretability. Based on original fieldwork, I show that Balti instantiates exactly this predicted configuration, providing a striking case in which a theoretically necessary type is empirically discovered and statistically validated. This study documents a relatively under-explored configuration, offering insights into subjects, pivots, agreement, and clause structure

1.9.5 A Unifying Theoretical Framework with Quantitative Validation: The study demonstrates that the cluster of properties — Phi_Availability, Rich_Agreement, Morph_ProDrop, and No_Expletive — are not merely correlated but computationally linked reflexes of a single underlying dimension, φ -state: predictable consequences of having $[u\varphi]$, $[i\varphi]$, or $[n\varphi]$. This provides a principled, causal explanation for why these properties co-vary cross-linguistically.

I employ Principal Component Analysis on an eighty-two-language sample, providing systematic statistical evidence that φ -state constitutes the primary dimension (PC1: 96.60% variance explained), with Phi_Availability tied as the strongest-loading indicator, alongside Morph_ProDrop and No_Expletive (25.61% each), structuring morphosyntactic variation with PC1 and PC2 together accounting for 100% of the structured morphosyntactic variance. This internal statistical structure is independently corroborated by an external test: φ -Availability values derived mechanically from raw Universal Dependencies treebank morphology — a corpus built without any reference to MPH — recover the hand-coded classifications at 95–97.5% accuracy (§1.6.4), with every residual discrepancy independently explained by the theory's own finer distinctions rather than left as noise. The convergence of statistical structure, independent corpus validation, and theoretical prediction offers strong empirical support for the unified model.

Taken together, these contributions establish the Morphosyntactic Pivot Hypothesis (MPH) as:

- an operational, testable, statistically validated, and reproducible minimalist model, **independently corroborated by corpus data external to the theory**,
- consistent with a configuration predicted by the theory but not clearly documented before,
- a demonstration of a compact parameter set accounting for the full extent (100%) of observed structured variance

2. Summary of Cross-Linguistic Findings

This section summarizes the outcomes of the two-phase identification procedure—syntactic nomination followed by morphological corroboration—for each language. Full diagnostic inventories, sample sentences, glosses, judgements and quantitative results are provided in Appendix E (§§1–6). Here I highlight only the convergent evidence identifying the morphosyntactic pivot in each system.

2.1 English: T as the Morphosyntactic Pivot

Syntactic identification.

In English, clause-boundary diagnostics consistently point to T(ense) as the clause-defining head. First, coordination reveals that TP is the natural unit: finite clauses coordinate only if both conjuncts bear tense (*John left and Mary arrived*), whereas aspectual or agreement morphology alone does not license coordination. Second, complementizer selection confirms the same: the complementizer *that* uniformly selects a TP, yielding embedded finite clauses with tense specification (*I know that John*

left). Third, ellipsis patterns show that auxiliaries associated with T survive as remnants in fragment answers (*Did John leave? – Yes, he did*), indicating that TP material anchors clause-sized ellipsis (see Appendix E.1.1.1)

Movement diagnostics further corroborate T as the pivot. Subjects obligatorily raise to Spec-TP, as shown by adverb ordering and binding asymmetries (subjects surface above VP-internal adverbs and their binding domains). Extraction asymmetries also implicate T: subject movement interacts with Wh- and topicalization processes in ways consistent with an intermediate stop in Spec-TP. Finally, freezing effects confirm this configuration: extraction from within a subject that has raised to Spec-TP is degraded, in line with predictions if T hosts the EPP feature. Taken together, these tests nominate T as the syntactic candidate pivot in English (see Appendix E.1.1.2).

Morphological corroboration.

Bybee's (1985) criteria provide partial but sufficient support to this nomination. Tense marking in English is obligatory in finite clauses: every finite verb form requires specification for tense (past vs non-past), typically realized on the auxiliary or main verb. Paradigmatic richness is comparatively reduced — contrasts are largely restricted to past vs. non-past and the 3sg present marker -s — but these distinctions are stable. Productivity is high: tense marking extends across the verbal lexicon and applies to novel forms without restriction (see Appendix E.1.2).

Verdict

English, thus, illustrates a system where both syntactic diagnostics and morphological criteria converge on T as the morphosyntactic pivot, despite the relative poverty of the tense paradigm (see Appendix E.1.3).

2.2 Hindi-Urdu: Asp as the Morphosyntactic Pivot

Syntactic identification.

In Hindi-Urdu, clause-boundary diagnostics clearly implicate Asp rather than T as the pivot. First, coordination is possible at the level of AspP: conjuncts headed by perfective or progressive aspect can form full clauses even without an overt tense auxiliary (*Abdullah-ne roti khā-ī aur kofī pi-ī* 'Abdullah ate bread and drank coffee'). Second, complementizer selection supports this view: the complementizer *ki* ('that') can embed clauses that contain aspectual marking but lack tense morphology (*Main chāhtā hū [ki Abdullah roti khā-ī]* 'I want that Abdullah eat bread'). Third, ellipsis confirms Asp's clausal status: in fragment answers, it is aspectual morphology that survives (*Khā-ī* 'ate'), while tense morphology may be absent (see Appendix E.2.1.1).

Movement diagnostics converge on the same conclusion. Subjects surface in Spec-AspP, not Spec-TP. Ergative subjects in perfective clauses consistently precede aspectual morphology, showing that they have moved to a position defined by Asp. Wh-subjects also pass through AspP en route to CP, with adverb placement and intervention effects revealing an intermediate landing site in Spec-AspP. Finally, freezing effects arise once subjects occupy Spec-AspP: extraction from within such subjects is degraded, consistent with Asp hosting the EPP-related subject position. Collectively, these diagnostics nominate Asp as the syntactic candidate pivot (see Appendix E.2.1.2).

Morphological corroboration.

Aspectual morphology is obligatory in finite clauses: a clause cannot be complete without specification for perfective, progressive, or habitual. The system is paradigmatically rich, marking not only aspectual contrasts but also gender and number agreement on participles. Critically, Asp bears [iφ] in both imperfective and perfective contexts — the same functional head — but ergativity in perfective clauses blocks subject-agreement while permitting object-agreement to surface (see Appendix E.2.2).

Verdict.

The convergence of syntactic diagnostics and morphological criteria establishes Asp as the morphosyntactic pivot in Hindi–Urdu. This confirms that clause-defining functions can be anchored in heads other than T and demonstrates how the syntax-first-morphology-second approach reduces the risks of circularity (see Appendix E.2.3).

2.3 Swahili: Agr as the Morphosyntactic Pivot

Syntactic identification.

In Swahili, independent syntactic diagnostics point to Agr (agreement morphology on the verb) as the clausal pivot. Clause-boundary tests show that finite clauses require subject agreement prefixes: conjuncts cannot coordinate unless both carry agreement, and embedded clauses introduced by complementizers still obligatorily display agreement. Ellipsis further supports this: in fragment answers, the agreement prefix suffices to license a null subject (*Ali-enda?* – *Ndiyo, a-li-enda* ‘Did he go? – Yes, he went’). Agreement morphology thus acts as the defining marker of clausehood. (see Appendix E.3.1.1).

Movement diagnostics reinforce this conclusion. Subject licensing in Swahili is tied to agreement morphology: pro-drop is possible only when the subject agreement prefix is present, indicating that the subject’s landing and licensing site is associated with Agr. Extraction asymmetries further implicate agreement: Wh-extraction and topicalization interact with agreement in ways that show Agr is an intervener in movement paths. Freezing effects are also consistent with this analysis: once a subject is licensed via Agr, internal extraction from it is degraded. Collectively, these diagnostics nominate Agr as the syntactic candidate pivot (see Appendix E.3.1.2).

Morphological corroboration.

Bybee’s criteria strongly confirm this nomination. Agreement morphology is obligatory in all finite clauses, with no exceptions. It is paradigmatically rich, marking person/number distinctions alongside an extensive noun-class system. It is also systematically productive, attaching to virtually all verbs in the language and extending across clause types. Unlike tense or aspect markers, agreement morphology alone suffices to signal finiteness (see Appendix E.3.2).

Verdict.

The convergence of syntactic and morphological evidence identifies Agr as the morphosyntactic pivot in Swahili. This aligns with the pro-drop pattern and highlights a system in which subject licensing is rooted in agreement, not tense or aspect (see Appendix E.3.3).

2.4 Italian: Mood as the Morphosyntactic Pivot

Italian provides a revealing test case for the Morphosyntactic Pivot Hypothesis, representing a stage in the evolution of clause structure. Whereas Latin maintained a strict division between Tense and Mood morphology, it has developed over a period a high degree of morphological fusion between the two while retaining syntactic independence.

Syntactic Identification

The standard six syntactic diagnostics—coordination, complementizer selection, ellipsis, subject-position tests, extraction, and freezing effects—uniformly point to Mood as the locus of clausal organization. Clauses marked for Mood behave as complete finite units, and subjects are licensed in the structural domain associated with Mood rather than Tense.

To verify this preliminary result, a series of seven additional diagnostics were applied, targeting phenomena that are particularly sensitive to clause-internal feature hierarchies: polarity licensing, temporal-dependency patterns, scope relations, particle distribution, ellipsis asymmetries, subject-position contrasts, and, most decisively, the clitic-placement test. Together these diagnostics provide converging evidence that Mood, not Tense, defines the finite clause in Italian. Across all tests, mood variation categorically affects grammaticality (100% → 0% acceptance), whereas tense variation within a given mood produces no syntactic contrast (see Appendix E.4.1.1 & E.4.1.2).

Morphological Corroboration

Despite pervasive fusion of Mood and Tense morphology, the morphological data corroborate the syntactic conclusion. Mood specification is obligatory in all finite clauses: every finite verb overtly marks one of the major moods (indicative, subjunctive, conditional, or imperative), and these paradigms are fully productive across verb classes. In contexts where partial separation is observable—imperatives (Mood without Tense) and subjunctives (Mood-determined temporal interpretation)—Mood remains obligatory in three of the four clause types, whereas Tense is obligatorily realized in only one. This asymmetry confirms that the Italian clause is anchored in Mood (see Appendix E.4.2).

Verdict and Diachronic Interpretation

The convergence of thirteen syntactic diagnostics, including the decisive clitic-placement test, with robust morphological obligatoriness establishes Mood as the morphosyntactic pivot in Italian. Italian thus aligns with Swahili (Agr + [iφ]) and Hindi-Urdu (Asp + [iφ]) as a pivot-switch language whose interpretable φ-features license pro-drop, contrasting with English (T + [uφ], no pro-drop). The persistent Mood-Tense fusion cannot be resolved synchronically; instead, it marks a distinct diachronic phase in the language's evolution—a stage where syntactic reorganization (anchoring the EPP in Mood) has already occurred while morphological transparency lags. This fusion, therefore, reflects historical layering rather than analytic indeterminacy, capturing the shift from the T-based clause of Latin to the Mood-based architecture of modern Italian (see Appendix E.4.3).

2.5 Balti: Asp/Evid as the Morphosyntactic Pivot

Syntactic identification

In Balti, clause-boundary diagnostics consistently nominate Asp/Evid as the pivot head without any φ features. Coordination demonstrates that clauses can conjoin around evidential or aspectual auxiliaries (*Na-la aasho thong-sad pa dang go-sad* 'I saw the woman and (I) went'), even in the absence of an independent tense marker.

Complementizer selection yields the same result: complementizers like *kih* embed clauses whose finiteness is guaranteed by evidential auxiliaries (*Na la shespo kih kho dy aasho la hlataen yod* 'I think that he/she is seeing the woman'). Ellipsis further confirms this role: in fragment answers, the evidential auxiliary (*yod, yin, song, zer*) survives, showing that Asp/EvidP alone anchors clausehood (see Appendix E.5.1.1).

Movement diagnostics reinforce this identification. Subjects surface in Spec-Asp/EvidP, not Spec-TP. Ergative subjects in perfective clauses obligatorily appear above the auxiliary (*Na-la aasho thong-song* 'I saw the woman'), revealing Asp/Evid as the landing site. Wh-subjects also pass through this position en route to CP, as shown by their placement relative to adverbs (*Su-la aasho thong-sad?* 'Who saw the woman?'). Freezing effects arise when subjects occupy Spec-Asp/EvidP: extraction from within such subjects is degraded, consistent with the projection hosting the EPP bundle. Collectively,

these diagnostics nominate Asp/Evid as the syntactic candidate pivot (details of judgments from N=30 Balti speakers are in Appendix E.5.1.2).

Morphological corroboration.

Bybee's criteria confirm this nomination. Evidential and aspectual auxiliaries are obligatory in finite clauses, regardless of tense specification. The system is paradigmatically rich, distinguishing perfective, progressive, evidential, and epistemic forms through auxiliary alternations (*yod, yin, song, zer*). It is also systematically productive, applying across the full verbal lexicon and extending to novel predicates. By contrast, tense markers alone are insufficient to define clausehood (see Appendix E.5.2).

Verdict

The convergence of syntactic and morphological evidence establishes Asp/Evid as the morphosyntactic pivot in Balti. This makes Balti typologically parallel to Hindi–Urdu, but with the additional feature that evidentiality plays a central role in defining clause structure (see Appendix E.5.3).

2.6 Warlpiri: Agr as the Morphosyntactic Pivot

Note on data provenance: This analysis relies on published descriptions (Hale 1973, 1983; Simpson 1991; Legate 2002, 2008; Laughren 2002; Bower 2014) and does not include primary fieldwork. Claims are accordingly restricted to what can be established from the cited literature and existing corpora.

Syntactic identification.

In Warlpiri, clause-boundary diagnostics point to Agr (the auxiliary clitic complex) as the clause-defining head. First, coordination shows that clauses require the auxiliary complex: coordinated clauses each bear auxiliaries encoding subject and object agreement (*Ngarrka-ngku ka karli jarnti-rni manu pama-rnu* 'The man is trimming a boomerang and speared it'), while coordination without auxiliaries is ungrammatical. Second, complementizer selection confirms this: complementizers select clauses containing the auxiliary complex, which encodes agreement and occupies the second position (Wackernagel position) in the clause. Third, ellipsis patterns show that the auxiliary can survive as a minimal fragment, anchoring clause structure independently of the main verb (see Appendix E.6.1.1).

Movement diagnostics reveal a distinctive pattern. Warlpiri exhibits radical free word order for lexical DPs, but the auxiliary complex occupies a fixed second position. This auxiliary hosts subject and object agreement morphology and satisfies the EPP not through DP movement to its specifier, but through head-movement of the auxiliary itself to the clausal second position. Extraction processes interact with the auxiliary position, showing intervention effects consistent with the auxiliary as a phase boundary. The distinctive property is that subjects (and other arguments) remain in or near vP, with agreement features checked via the auxiliary that has moved to second position. Collectively, these diagnostics nominate Agr as the syntactic candidate pivot, though the system differs from other pivots in satisfying the EPP through head-movement rather than phrasal movement to Spec-PivotP (see Appendix E.6.1.2).

On the status of second position: The analysis claims Warlpiri auxiliary movement to second position satisfies the EPP, raising the question of whether second position is a syntactic position or merely a PF (Wackernagel) requirement. The evidence suggests both: second position is syntactically fixed AND constitutes the locus of EPP satisfaction.

Evidence for syntactic status:

Complementizers occupy second position: The complementizer *kuja* 'that' appears in second position in embedded clauses, suggesting second position is a

syntactic slot (like C° or a high functional head) rather than purely prosodic placement.

Structural fixedness: Auxiliaries cannot appear clause-finally or in other positions regardless of prosodic or information-structural factors. This rigidity indicates syntactic positioning rather than variable PF adjustment.

Extraction interaction: Long-distance wh-extraction shows that movement interacts with auxiliary position systematically—the auxiliary marks phase boundaries that extraction must transit. This suggests the auxiliary occupies a syntactically active position mediating A'-dependencies.

Agreement obligatoriness: The agreement morphology on the auxiliary is obligatorily realized and cannot be omitted even when DPs are overt. This syntactic obligatoriness indicates the auxiliary position itself is syntactically required, not merely phonologically convenient.

The analysis thus treats second position as a syntactic projection where the auxiliary (Agr) head moves to satisfy [uEPP]. This movement creates an obligatory functional projection in a structurally fixed location—parallel to DP movement to Spec-TP in configurational languages but realized through head-movement rather than phrasal movement. Warlpiri demonstrates that the EPP requirement (obligatory structural anchoring) can be satisfied through different movement strategies cross-linguistically (see Appendix E.6.1.1 and E.6.1.1.2)

Morphological corroboration.

The auxiliary complex in Warlpiri is obligatory in all finite clauses, appearing in over 95% of matrix clauses in corpus data (Bowern 2014). It is paradigmatically rich, distinguishing person and number for both subject and object across an extensive paradigm (1/2/3 × SG/DU/PL for subjects; similar distinctions for objects). The system is productive, applying to novel predicates, and general, extending across all verb classes without exception. By contrast, tense/aspect morphology appears on the main verb but is optional in many contexts and less paradigmatically elaborate (see Appendix E.6.2).

Verdict

Syntactic diagnostics strongly identify the auxiliary (Agr complex) as the pivot, with morphological evidence providing robust corroboration. Warlpiri thus exemplifies an Agr-pivot system where the pivot bears [iφ] for both subject and object, licensing pro-drop for both argument positions. The distinctive feature is that the EPP is satisfied through pivot-head-movement to second position rather than through DP-movement to Spec-PivotP. This makes Warlpiri typologically parallel to Swahili (Agr-pivot with [iφ] and pro-drop) but with a unique EPP-satisfaction strategy.

Theoretical significance: Warlpiri demonstrates that EPP satisfaction need not involve phrasal movement to Spec-PivotP. Instead, the pivot itself moves (head-movement to second position), satisfying both the second-position requirement and the EPP's demand for an obligatory projection. This dissociates the EPP from "subject raising" while maintaining that the EPP is hosted on the morphosyntactic pivot. The analysis supports the broader MPH claim: the pivot is identified by structural centrality (second position is structurally fixed; DPs are radically free), morphological prominence (auxiliary is obligatory and paradigmatically rich), and EPP-hosting—but EPP satisfaction can proceed through head-movement rather than specifier-filling. Warlpiri is thus an Agr-pivot language with [iφ], licensing robust pro-drop for both subject and object. (see Appendix E.6.3).

2.7 The EPP Hosting Maxim

Across all languages investigated, the locus of the EPP-bundle—[uEPP], [iCase], and ($\pm\varphi$)—coincides with the morphosyntactic pivot identified through the Biphasic identification diagnostics. The empirical pattern generalizes as follows:

EPP-Pivot Convergence (Empirical Generalization)

The functional head that satisfies independent syntactic diagnostics of structural centrality and meets morphological criteria of prominence (obligatoriness, paradigmatic richness, relevance, and generality) is the unique host of the EPP-bundle. This head obligatorily projects, licenses case via [iCase], and anchors clause structure by valuing [uEPP]. Its φ -state is an indicator of the presence and interpretability of the φ -feature, rich agreement, morphological pro-drop and expletives.

Theoretical significance: This convergence between syntactic structural centrality and morphological prominence is unexpected under theories treating syntax and morphology as loosely coupled. However, it follows if obligatory projection and morphological robustness target the same grammatical position, potentially for learnability reasons (surface morphology makes the pivot identifiable) or architectural economy (a single head hosts all finite-clause requirements).

The EPP-Pivot Convergence captures the cross-linguistic observation that EPP-hosting is a derived property of pivothood, not an *a priori* universal of T. Once the pivot is independently identified, the distribution of EPP effects follows predictably from the pivot's feature composition.

Crucially, this generalization leads to the Hierarchical Parametric Organization of the locus of the EPP itself, requiring two distinct modes of Agree. In languages whose pivots bear [i φ] or [u φ], EPP satisfaction proceeds through either Unimodal Agree (valuation of [uEPP] and [uCase] with or without φ -features) or Bimodal Agree (matching for φ -features and valuation for [uEPP] and subject's [uCase]).

Section 3 develops this theoretical consequence formally in terms of the φ -Availability and φ -Interpretability Parameters. This unified account explains cross-linguistic variation in case, agreement, pro-drop, and expletives as predictable outcomes of hierarchical structure of the EPP-hosting pivot as evidenced by the high concentration on PC1 (96.60%) and secondary structured variation on PC2 (3.40%).

3. The φ -Availability and the φ -Interpretability Parameters: Hierarchical Parametric Organization

3.1 Motivation: The Hierarchical Parametric Structure

The cross-linguistic diagnostics in Section 2 reveal that the morphosyntactic pivot varies systematically in its featural specification, with direct consequences for case licensing, subject realization, pro-drop, expletives, and agreement. This variation exhibits a hierarchical structure rather than flat typological diversity: languages do not select freely among three unrelated pivot types — 0.1 (*n φ*), 0.2 (*u φ*), 1.0 (*i φ*) — but make parametric choices at two distinct levels that interact predictably.

Following the terminology of latent-variable measurement theory, a language's joint setting on the φ -Availability and φ -Interpretability Parameters is its **φ -state**. φ -state is not itself a directly observed property; it is a latent grammatical parameter, instantiated in the narrow syntax as the pivot's featural specification (§3.2), detectable empirically only through four independently testable surface reflexes: the composite φ -state at LF (*Phi_Availability*), the richness of verbal agreement (*Rich_Agreement*), the availability of morphological pro-drop (*Morph_ProDrop*), and the licensing status of expletive subjects (*No_Expletive*). Each reflex is generated by φ -state but does not itself define it — φ -state is not a composite or index of the four properties, but their common cause. This

reflective structure licenses treating convergence among the diagnostics (§3.5) as evidence for φ -state's existence rather than an artifact of how the diagnostics were built. That claim is tested in this study by two independent means: internal statistical structure across a genealogically diverse 82-language sample (§3.5), and external corroboration from Universal Dependencies treebank data — a corpus assembled without reference to MPH (§3.5.5). Both converge on the same latent structure, which is the strongest evidence available that φ -state is a real organizing property of grammar rather than a descriptive convenience.

A language's φ -state takes one of three values — 0.1 (φ -unavailable), 0.2 (φ -available, uninterpretable), 1.0 (φ -available, interpretable) — corresponding to Types 1–3 (§3.3). All φ -less systems uniformly lack person-number morphology on the pivot, require discourse context for null-subject licensing, lack expletives, and show aspectual or evidential morphology as the obligatory clause-defining marker. This clustering follows from hierarchical parametric interaction, not from independent parametric choices operating simultaneously.

3.2 The φ -Availability and Interpretability Parameters

3.2.1 The φ -Availability Parameter

The φ -Availability Parameter specifies whether a language activates φ -features at the pivot head at all — an ontological question, logically prior to any question of how those features are computed.

In φ -available systems, the pivot bears φ -features — interpretable or uninterpretable — and establishes φ -dependencies with nominal arguments. English, Hindi-Urdu, Swahili, Italian, and Warlpiri instantiate this setting. Finite clauses require person-number/gender/class specification somewhere in the verbal morphology, whether impoverished (English third-singular -s) or elaborate (Swahili subject prefixes distinguishing eighteen noun classes). The pivot probes to value or match subject DPs, and case licensing proceeds in tandem with φ -agreement.

In [$n\varphi$] systems, the pivot lacks φ -features entirely; clause anchoring operates through aspectual or evidential specification instead. Balti instantiates this setting: finite clauses require perfective/progressive, direct/inferential, or egophoric/non-egophoric marking, but show no person or number distinction whatsoever. The pivot projects obligatorily and licenses case, satisfying the EPP through aspectual anchoring with no φ -checking; subjects remain visible to the computational system through their own interpretable φ -features but never enter a dependency with the pivot.

φ -available systems should show person-number/gender paradigms on at least some verbal forms and permit either obligatory overt subjects or pro-drop, depending on interpretability. φ -less systems should uniformly lack such paradigms, show aspectual/evidential paradigms instead, permit pro-drop only when discourse context supplies recoverability, and show case systems varying independently of any φ -specification.

3.2.2 The φ -Interpretability Parameter

Once φ -features are activated, their interpretability determines how φ -relations are realized. This parameter operates only within φ -available systems - a computational question about the treatment of a φ -specification already established to exist, not about its existence.

Uninterpretable φ -features trigger valuation-based **Unimodal Agree**. English T bears [$u\varphi$], probing for and valuing against a goal's [$i\varphi$], then deleting [$u\varphi$] to satisfy Full Interpretation. [i Case] on T values [u Case] on the goal as nominative in the same dependency. Because uninterpretable φ requires a valuator, the derivation needs an

overt DP or expletive insertion when no thematic subject is available. Post-valuation deletion means ϕ -information does not persist morphologically — blocking pro-drop. **Interpretable ϕ -features** trigger matching-based **Bimodal Agree**. Hindi-Urdu, Swahili, Italian, and Warlpiri instantiate this setting: pivot and argument both bear semantically contentful ϕ that must be compatible rather than one valuing the other. Because both feature sets are interpretable, they persist to both LF and PF; at PF they surface as agreement morphology recoverable even when the argument is phonologically null, making pro-drop available.

A critical refinement, established empirically in §3.5.3 and §3.8: the "cascade of consequences" from interpretability is not uniform across all four reflexes.

Phi_Availability, Morph_ProDrop, and No_Expletive track interpretability deterministically ($r=1.000$ across the full sample). Rich_Agreement does not — it can survive diachronically after a pivot's ϕ -features have already become uninterpretable, producing the Type 2b languages (German, Icelandic, Faroese, Finnish) discussed throughout this section. The theoretical claim is therefore precise, not approximate: interpretability determines subject-licensing behavior categorically, while it determines agreement morphology only as a tendency subject to independently-motivated diachronic exceptions.

3.3 The Hierarchical Parametric Organization

ϕ -availability and ϕ -interpretability enter a *logical-dependency relation*: interpretability is defined only if ϕ -features are available at all. This is not merely a technical dependency but reflects a deeper asymmetry in *kind* between the two parameters. ϕ -Availability is an **ontological** parameter: it asks whether ϕ -features exist in the derivation at all — whether the pivot bears any ϕ -specification for the computational system to operate on. ϕ -Interpretability is a **computational** parameter: it asks how an already-existing ϕ -specification is treated by the grammar — whether it is valued and deleted (Unimodal Agree) or matched and retained to LF (Bimodal Agree). Existence is a precondition for computation: a feature must be present in the derivation before any operation — valuation, matching, deletion — can apply to it. No procedure can compute the interpretability of a feature that is not there. Hierarchical dependency therefore follows not from a stipulated ordering convenient for reducing four logical cells to three, but from the logical entailment that computation presupposes an object to compute over. ϕ -availability is therefore not merely prior by convention but ontologically and logically prior; ϕ -interpretability is a secondary, computational refinement operating only within ϕ -available systems. This dependency yields three macro-types, not four: [- ϕ , $\pm$ interp] is incoherent, since absent features cannot be interpretable or uninterpretable.

The ϕ -Interpretability Parameter [$\pm$ interp] (contingent on [$+\phi$]) asks: do ϕ -features on the pivot provide semantically contentful material to LF?

- [-interp] → **Type 2 ([$u\phi$])**: ϕ -features lack semantic content at LF despite morphological presence (e.g., English residual agreement, or German/Icelandic/Faroese/Finnish's fully productive agreement)
- [+interp] → **Type 3 ([$i\phi$])**: ϕ -features carry semantic content reaching LF (e.g., Italian, Hindi-Urdu, Swahili)

The three resulting types:

1. **Type 1 ($n\phi$)**: aspectual/evidential pivots, no person-number/gender inflection, discourse-based pro-drop, no expletives.

2. **Type 2 (ϕ -available + [$u\phi$]):** valuation-based pivots, obligatory subjects and expletives, agreement richness independently variable (Type 2a: impoverished; Type 2b: rich but non-interpretive).
3. **Type 3 (ϕ -available + [$i\phi$]):** matching-based Agr/Asp/Mood pivots, morphologically licensed pro-drop, rich agreement, no expletive.

The three-way typology follows *logically* from a binary-plus-dependent architecture, not from ad hoc tri-partitioning. Properties distinguishing ϕ -available from ϕ -less systems — morphology, case alternations, pivot-mediated subject licensing — should strongly correlate across languages; §3.5 tests this directly.

Diachronically and in acquisition, ϕ -availability is the more fundamental, earlier-acquired, and more change-resistant choice; ϕ -interpretability may shift within ϕ -available languages through erosion or reanalysis without loss of ϕ -based finiteness (exactly the Type 2b pattern; §3.8). Parametric dependency correspondingly reduces the acquisition search space: if ϕ -Availability = [–Available], ϕ -Interpretability need not be evaluated at all. This predicts staged acquisition, with primary-parameter values converged on before secondary values become detectable.

3.4 Bimodal Agree: Formal Characterization

The two-level architecture determines how Agree proceeds, yielding fundamentally different licensing mechanisms for ϕ -available and ϕ -less systems.

In ϕ -available systems, the pivot bears [$i\phi$] or [$u\phi$] and establishes ϕ -dependencies with arguments bearing [$i\phi$]. Bimodal Agree involves matching of ϕ -features with valuation of [$u\text{Case}$]/[$u\text{EPP}$]; Unimodal Agree values [$u\text{Case}$]/[$u\text{EPP}$] while ϕ -features are either valued or absent. Case licensing proceeds through the same operation as ϕ -checking; subject position is determined by the need to satisfy both ϕ -agreement and EPP, yielding specifier-movement in most systems or head-movement of the pivot itself, as in Warlpiri.

Compatibility between probe and goal requires overlapping ϕ -specifications.

Incompatibility arises from feature deficit (goal lacks ϕ entirely), feature clash (probe's specification finds no match), or intervention. In valuation mode, the probe's [$u\phi$] are valued against the goal's [$i\phi$] and deleted from the narrow syntax. In matching mode, the probe's [$i\phi$] and the goal's [$i\phi$] are co-indexed and both persist to the interfaces.

In [$n\phi$] systems, Agree proceeds without ϕ -interaction at all: the pivot bears only [$i\text{Case}$] and [$u\text{EPP}$], values the argument's [$u\text{Case}$] directly, and satisfies [$u\text{EPP}$] through obligatory aspectual/evidential projection. The argument's ϕ -features remain unprobed, persisting to LF as freestanding features recoverable only through discourse. This bifurcation is structural, not merely descriptive: the presence or absence of ϕ -features on the pivot determines whether Agree can proceed through ϕ -mediation at all, yielding two qualitatively different derivations that nonetheless satisfy the same universal requirement — some functional head must anchor case and EPP.

3.5 Quantitative Validation of the Hierarchical Parametric Organization

3.5.1 Three-Sample Validation Strategy

To test whether ϕ -state organizes morphosyntactic variance as predicted, I employed a three-sample validation strategy assessing both internal consistency and generalizability:

Sample 1: Core Languages (N=7) — Intensive documentation of maximally diverse languages (Balti, English, Hindi, Urdu, Italian, Swahili, Warlpiri) representing all three parametric types across seven language families/subfamilies and five continents, each tested with twelve syntactic probes.

Sample 2: Outgroup Languages (N=75) — Systematic expansion for genealogical diversity and typological coverage, testing generalization to broader typological space including edge cases and mixed systems.

Sample 3: Combined Sample (N=82) — Unified analysis validating that core and outgroup patterns reflect the same underlying dimension rather than sample-specific artifacts.

Methodological triangulation: each sample was analyzed via (1) Principal Component Analysis, (2) polychoric correlation analysis, and (3) Multiple Correspondence Analysis. A fourth, independent layer of evidence — corpus-derived validation via Universal Dependencies — is introduced in §3.5.5, providing a test of the theory's predictions using data the theory played no role in generating.

3.5.2 Primary Result: High Dimensional Concentration

PCA on four variables (Phi_Availability, Rich_Agreement, Morph_ProDrop, No_Expletive) yields:

- PC1 = **96.60%** (eigenvalue = 3.864)
- PC2 = **3.40%** (eigenvalue = 0.136)
- Cumulative = **100.00%**

The concentrated **96.60:3.40** variance ratio validates MPH's prediction: a primary dimension (ϕ -state) dominates morphosyntactic organization, relegating secondary factors to negligible residual status.

Variable contributions to PC1: Phi_Availability, Morph_ProDrop, and No_Expletive contribute equally (25.61% each), confirming deterministic bundling. Rich_Agreement contributes 23.17% — lower, reflecting the theoretically important Type 2b exceptions (German, Icelandic, Faroese, Finnish: rich morphology, uninterpretable ϕ).

PC2 (3.40%) captures the residual: Rich_Agreement dominates it (76.83% contribution), distinguishing Type 2a (English, Dutch, French: degraded morphology) from Type 2b (German, Icelandic, Faroese, Finnish: robust paradigms) within the same $[u\phi]$ category. The small magnitude confirms this within-type variation is subordinate to the primary parametric-type distinction.

3.5.3 Perfect and Near-Perfect Correlations

MPH predicts deterministic bundling for the core diagnostics. Under the reflective model (§3.1), these are convergent evidence that all four indicators track a shared latent ϕ -state, not pairwise causal relationships among the indicators themselves.

- **Phi_Availability $\leftrightarrow$ Morph_ProDrop: $r = 1.000$** (exact, 82/82 languages)
- **Phi_Availability $\leftrightarrow$ No_Expletive: $r = 1.000$** (exact, 82/82 languages)
- **Phi_Availability $\leftrightarrow$ Rich_Agreement: $r = 0.908$** (strong but, as predicted, imperfect — Type 2b exceptions)

Polychoric correlation analysis — appropriate for the underlying ordinal structure — returns all pairwise coefficients at or near 1.000, including Rich_Agreement. This reflects a genuine empirical fact (near-total categorical determinism in the corrected 82-language sample: essentially no counter-examples remain in any pairwise contingency table) but is reported here with an important statistical caveat: at this level of determinism, polychoric maximum-likelihood estimation is a boundary solution rather than an interior estimate, and should be interpreted as *"no counter-examples were observed across 82 languages for any pairwise combination,"* not as an independently informative correlation coefficient distinct from the Pearson values above.

3.5.4 Methodological Robustness

Cross-sample stability:

- Core languages (N=7): PC1 = 99.91%, PC2 = 0.09%

- Outgroup (N=75): PC1 = 96.34%, PC2 = 3.66%
- Combined (N=82): PC1 = 96.60%, PC2 = 3.40%

The 0.26-percentage-point gap between Outgroup-only and the full sample confirms the result is not an artifact of Core-language selection.

Leave-one-out jackknife analysis (82 iterations) confirms exceptional stability: range [96.5%, 97.4%] (0.9% spread), mean 96.6% (matching the full-sample estimate), SD = 0.176% — all 82 iterations successful. Maximum influence is shared across German, Icelandic, Faroese, and Finnish (the Type 2b cluster), consistent with their theoretically-motivated boundary status rather than reflecting any single leverage point.

Multiple Correspondence Analysis provides independent validation under an unordered-categorical treatment: Dimension 1 = 55.45%, Dimension 2 = 42.86%, cumulative = 98.31%. A quadratic regression of Dim 2 on Dim 1 ($R^2 = 0.966$) confirms Dim 2 is a Guttman/horseshoe artifact of representing near-deterministic ordinal categories on orthogonal axes, not independent structure — MCA thus corroborates the same single-dimension ϕ -state identified by PCA.

Study	Languages	Features	Method	PC1
Bickel & Nichols (2005)	72	~20	Factor analysis	45%
Dunn et al. (2011)	301	125	PCA	52%
Current study	82	4	PCA	96.60%

A **51.60**-percentage-point improvement over Bickel & Nichols and a **44.60**-point improvement over Dunn et al. reflects two principled differences: (1) **theory-driven targeting** — diagnostics were explicitly predicted to bundle from interface-level parameter specifications, whereas prior studies aggregated heterogeneous features without parametric predictions; (2) **interface-level parameter** — ϕ -Interpretability operates at the syntax-semantics interface with cascading morphological, syntactic, and interpretive consequences, producing tighter clustering than purely syntactic parameters with localized effects.

3.5.5 Independent Corpus Validation via Universal Dependencies

The internal statistical structure above (§3.5.2–3.5.4) is, by construction, computed from hand-coded values. To guard against the possibility that this structure reflects coding artifacts rather than genuine linguistic organization, Phi_Availability was independently tested against raw morphological data from the Universal Dependencies treebank corpus (v2.17; 339 treebanks, 186 languages) — a resource built by hundreds of annotators over two decades with no relationship to MPH.

A corpus-derived ϕ -index — a mechanical, zero-learned-parameter count of Person/Number/Gender feature co-occurrence on finite verbs — was computed for 80 of the 82 sample languages (Swahili and Balti lack UD treebanks and were validated instead via primary elicitation; §1.6.1). Classification thresholds were fixed a priori and never fit to the gold labels.

Results: ϕ -availability (0.1 vs. {0.2, 1.0}) is recovered at 97.5% accuracy (AUC=0.998); ϕ -interpretability (0.2 vs. 1.0) at 95.0% accuracy (AUC=0.966). Critically, all four residual misclassifications are independently explicable rather than unexplained noise, and fall into two theoretically distinct categories:

- **Norwegian and Swedish** (false negatives on availability): both have completely lost verbal person-marking, morphologically indistinguishable from genuinely phi-less languages — yet the same syntactic diagnostics that identify German, English, and Dutch as [u ϕ] (obligatory subjects, obligatory expletives) correctly place them at 0.2, not 0.1.

- **Icelandic, German, and Finnish** (false positives on interpretability): these retain genuinely rich agreement (Rich_Agreement=1) — the opposite profile from Norwegian/Swedish — yet richness does not entail interpretability, the central empirical claim of Type 2b. This outcome is theoretically expected. φ -index quantifies the richness of overt agreement morphology, whereas φ -interpretability concerns the grammatical status of that morphology. Rich agreement is therefore necessary for φ -interpretability but is not sufficient to establish it.

A morphology-only measure cannot distinguish either case from Type 3, precisely because it lacks access to the pro-drop and expletive diagnostics that correctly classify all five languages regardless of whether their surface agreement is rich or entirely absent. This is not a shortfall of the corpus test; it is independent confirmation that φ -interpretability is doing genuine theoretical work beyond what morphological richness alone predicts.

This constitutes a second, independent line of evidence for φ -state, obtained from data the theory played no role in generating, using a threshold fixed before the test was run — precisely the kind of test a Popperian criterion of falsifiability demands (§1.9.2).

3.5.6 The Organization of the Locus of the EPP

Results support a hierarchical cascade: a language's φ -state functions as the generating parameter; when set to 1.0, its four reflexes — Phi_Availability, Rich_Agreement, Morph_ProDrop, No_Expletive — co-occur as predicted consequences, not as a causal chain. This explains both the tight PC1 clustering and the meaningful residual on PC2. The 96.60:3.40 PC1:PC2 ratio validates hierarchical dependency over independent parameters: if φ -availability and φ -interpretability varied independently, PC1 and PC2 would be roughly equal (~50:50). The observed ratio reflects logical dependency — availability is a gatekeeper parameter constraining, but not determining, interpretability.

Why PC2 is only 3.40%: Type 1 languages are uniformly [-available], so availability-internal variation appears only within Types 2 and 3. PC2 thus captures (1) the Type 2a/2b distinction, (2) residual Type 2/Type 3 separation not captured by PC1, and (3) morphological richness gradients within types — all subordinate to the primary parametric distinction, consistent with gatekeeper architecture.

3.5.7 Three Discrete Typological Clusters

Languages form three discrete clusters corresponding to MPH types:

Type 1 [φ -less] (N=17, 20.7%): Chinese, Cantonese, Japanese, Korean, Thai, Vietnamese, Balti, Tagalog, Cebuano, Yoruba, Indonesian, Javanese, Khoekhoe, Coptic, Bambara, Haitian Creole, Malayalam, and others. No φ -agreement, discourse-licensed null arguments, no expletive requirement.

Type 2 [$u\varphi$] (N=12, 14.6%): Type 2a (English, Dutch, French, Danish, Norwegian, Swedish, and others) — impoverished agreement; Type 2b (German, Icelandic, Faroese, Finnish) — rich agreement despite uninterpretable status. No pro-drop, obligatory expletives across both subtypes.

Type 3 [$i\varphi$] (N=53, 64.6%): Italian, Spanish, Russian, Hindi, Urdu, Swahili, Greek, Polish, Arabic, Basque, Georgian, and many others. Uniformly show pro-drop, no expletives, rich agreement.

Effect sizes confirm discrete clustering rather than continuous variation: Type 1 vs. Type 2, $d=2.18$ (large); Type 1 vs. Type 3, Cohen's d is **undefined** — both groups show zero within-type variance on PC1, a stronger result than any finite large effect size, reflecting complete internal homogeneity; Type 2 vs. Type 3, $d=16.96$ (very large).

3.6 Pro-Drop: Two Licensing Mechanisms

The two-level architecture predicts two distinct pro-drop mechanisms, tracking φ -availability, with φ -interpretability further differentiating outcomes within φ -available systems.

Pivot-mediated pro-drop occurs in $[i\varphi]$ systems (Swahili, Italian, Hindi-Urdu, Warlpiri): interpretable φ -features persist to PF as agreement morphology, making subject reference recoverable without an overt DP. This mechanism is discourse-independent — pro-drop applies regardless of topic continuity, shows no person-hierarchy asymmetries, and permits null subjects to introduce new referents, since recoverability is morphological rather than pragmatic.

Discourse-mediated pro-drop occurs in $[n\varphi]$ systems (Balti): the pivot lacks φ -features entirely, so φ -information is never probed and remains recoverable only pragmatically. Balti *thong-yod* ('is seeing') specifies no person or number; the null subject is licensed by discourse salience alone. This mechanism shows the opposite signature: strong discourse sensitivity, a person hierarchy favoring egophoric contexts, and resistance to introducing new referents via null subjects.

$[u\varphi]$ systems permit neither mechanism. In English, T's uninterpretable φ must be valued by an overt goal and is deleted post-valuation — blocking pivot-mediated pro-drop (no recoverable morphology survives to PF) and discourse-mediated pro-drop alike ($u\varphi$ still forces syntactic probing, requiring an overt subject or expletive). This blocking holds uniformly across $[u\varphi]$ systems regardless of agreement richness: German (Type 2b) blocks pro-drop despite rich morphology exactly as English (Type 2a) does with impoverished morphology, since the blocking factor is interpretability, not paradigm size — the clearest possible illustration that Rich_Agreement dissociates from the theory's other three diagnostics.

Both mechanisms, and their joint absence in $[u\varphi]$ systems, follow directly from the architecture without further stipulation, generating testable predictions about discourse sensitivity, person-hierarchy effects, and their dissociation from morphological richness.

3.7 Integration with Phase Theory

Phase theory provides the structural context within which the two-level architecture operates. The pivot performs all functions traditionally attributed to T under the Phase Impenetrability Condition (Chomsky 2000, 2001), functioning as the first phase-external head probing the vP edge, with featural specification determined by the φ -Availability and φ -Interpretability Parameters.

In φ -available $[u\varphi]$ systems, the phase-external pivot initiates downward search once vP transfers. English T, bearing $[u\varphi]$, $[iCase]$, $[uEPP]$, probes the transferred vP edge; $[u\varphi]$ is valued by the goal's $[i\varphi]$ and deleted; $[iCase]$ values $[uCase]$ as nominative; $[uEPP]$ attracts the DP to Spec-TP, or triggers expletive insertion when no thematic subject is available, since $[u\varphi]$ still requires a bearer even when semantically vacuous.

In φ -available $[i\varphi]$ systems, the pivot establishes matching relations at the vP edge instead. Swahili Agr, bearing $[i\varphi]$, $[iCase]$, $[uEPP]$, matches its interpretable φ with the subject DP's through co-indexation; both persist to the interfaces; $[iCase]$ values $[uCase]$; $[uEPP]$ is satisfied through the matching relation itself, triggering overt movement only as information structure requires. At PF, Agr's interpretable φ surfaces as subject prefixes, licensing null subjects.

Warlpiri instantiates a variant where the pivot originates *within* the vP phase and establishes matching relations with both subject and object before head-moving to second position — satisfying the EPP without phrasal movement to a specifier, and

explaining the coexistence of free DP order with a fixed auxiliary position. The moved auxiliary's interpretable ϕ licenses pro-drop for both arguments independently. In ϕ -less systems, the phase-external pivot probes for [uEPP] only. Balti Asp/Evid, bearing no ϕ but carrying [iCase] and [uEPP], values the subject's [uCase] directly as ergative or nominative depending on the pivot's own aspectual properties (perfective $\rightarrow$ ergative; non-perfective $\rightarrow$ nominative). No ϕ -probing occurs; [uEPP] is satisfied through obligatory aspectual/evidential projection; the subject's interpretable ϕ -features remain unprobed, available only for discourse-based interpretation. Phase theory thus supplies the computational framework within which ϕ -availability and ϕ -interpretability determine specific licensing mechanisms: the phase edge fixes where probing initiates, the pivot's featural specification fixes what feature-checking is possible, and interpretability fixes whether checking proceeds through valuation-with-deletion or matching-with-persistence.

3.8 Empirical Consequences and Typological Predictions

The two-level architecture generates specific, testable predictions about possible and impossible language types, property clustering, and diachronic trajectories — now tested against two independent evidence sources rather than one.

The framework predicts that certain property combinations should be unattested.

Languages should not show morphologically-licensed pro-drop while simultaneously requiring obligatory expletive insertion, since pivot-mediated pro-drop and expletive-obligatoriness diagnose interpretable and uninterpretable ϕ -features respectively and cannot co-occur on the same pivot. This prediction is exceptionless across the full 82-language sample (Phi_Availability, Morph_ProDrop, and No_Expletive are empirically inseparable, $r=1.000$; §3.5.3) and independently corroborated by the UD corpus test (§3.5.5). **Rich_Agreement is explicitly excluded from this inviolable core:** because agreement morphology can survive independently of interpretability through diachronic retention (§3.5.6, §5.5), rich person-number morphology may co-occur with expletive-obligatoriness and blocked pro-drop, as attested in German, Icelandic, Faroese, and Finnish. This is not a weakening of falsifiability but a more precise statement of it: the framework predicts deterministic bundling for three of four diagnostics and principled, independently-motivated dissociation for the fourth — and both halves of the prediction are borne out, on both statistical and corpus evidence. Languages should also not lack person-number morphology entirely while showing pivot-mediated pro-drop, nor show discourse-insensitive pro-drop without any agreement morphology. These predictions create falsifiable boundaries on cross-linguistic variation.

The framework predicts tight within-type clustering, evident in PC1's 96.60% concentration. ϕ -unavailable systems should uniformly lack person-number paradigms, show aspectual/evidential clause-defining morphology, exhibit aspect-conditioned (not ϕ -conditioned) case alternations, and permit discourse-mediated pro-drop with person-hierarchy effects. ϕ -available [$i\phi$] systems should show rich agreement, unrestricted pro-drop, no expletive requirement, and ϕ -interacting case systems. ϕ -available [$u\phi$] systems should require overt subjects or expletives, block pro-drop, and tie case licensing to ϕ -valuation — but agreement richness within this type is **not** uniform: Type 2a (English, Dutch, French) shows impoverished agreement; Type 2b (German, Icelandic, Faroese, Finnish) retains rich agreement despite uninterpretable status, confirming that agreement richness alone is not a reliable interpretability diagnostic. Clustering should be tighter within types than across them, and tighter on the three deterministic diagnostics than on Rich_Agreement specifically.

The framework predicts specific acquisition trajectories. Children acquiring $[i\varphi]$ systems should master agreement early, produce appropriate pro-drop from the first multiword stage, and show no expletive-omission stage. Deen's (2003) longitudinal study of Swahili acquisition confirms this: adult-like agreement and productive pro-drop from the onset of multi-clausal production, no expletive stage. Similar early mastery is reported for Italian and Spanish (Hyams 1986; Rizzi 1994; Guasti 2002).

Children acquiring $[u\varphi]$ systems should follow a different path, largely independent of Type 2a/2b status, since the relevant property is uninterpretability itself, not surface richness: an Optional Infinitive stage (Wexler 1998) before overt subjects are required, and an extended expletive-omission period. This is attested in English and, notably, in German — despite German's rich adult paradigm, child German shows an Optional Infinitive stage comparable to English, consistent with acquisition delay tracking interpretability rather than morphology.

Children acquiring $[n\varphi]$ systems should master aspectual/evidential markers early, produce discourse-based pro-drop from early stages, and may temporarily overgeneralize null subjects to discourse-inappropriate contexts before pragmatic licensing stabilizes (Deen 2003: §6).

The framework predicts asymmetric diachronic change. Shifts in φ -interpretability within φ -available systems (paradigm erosion or enrichment) should not require restructuring clause architecture; shifts between φ -available and φ -unavailable settings require more radical reanalysis. Surface changes in agreement richness should therefore occur more frequently than changes in fundamental clause-anchoring mechanism — a prediction directly confirmed by Type 2b's existence: German, Icelandic, Faroese, and Finnish retained ancestral agreement morphology after their pivots had already become uninterpretable, exactly the asymmetric-change pattern predicted here, independently visible in both the statistical clustering (§3.5.2) and the corpus validation (§3.5.5).

4. Comparison with the Existing Models

4.1 Upward Agree: Bjorkman & Zeijlstra 2019

Upward Agree proposes that agreement operations proceed via upward search (Bjorkman & Zeijlstra 2019, p. 488–492): goals bearing interpretable features $[iF]$ probe upward to value uninterpretable features $[uF]$ on c-commanding functional heads (Zeijlstra 2012, p. 497–500), reversing the directionality of Chomsky's (2000, 2001) downward Agree (Bjorkman & Zeijlstra 2019, p. 359–360). This eliminates stipulations about search direction and captures long-distance agreement phenomena where lower elements appear sensitive to higher functional specifications.

The fundamental challenge: Upward Agree maintains that φ -agreement universally involves $[u\varphi]$ on functional heads being valued by $[i\varphi]$ on DPs through upward search. The MPH, however, shows systems where pivots bear $i\varphi$ (not $u\varphi$) and establish matching relations, or are φ -less with no φ -probing whatsoever. These patterns pose challenges to Upward Agree's universal on-heads architecture.

Empirical Challenge 1: Balti φ -less Pivots

Balti aspectual-evidential pivots obligatorily project and license case without any φ -features:

- (1) *Go-ain* *yod*
 go-EGO EVID.NPST
 '(I) am going (I know directly) '

The aspectual-evidential auxiliary *yod* obligatorily projects and values ergative case on subjects, yet shows no person, number, or gender distinctions. The form remains invariant across all person-number-gender combinations.

The challenge for Upward Agree: Under Bjorkman & Zeijlstra's framework, when a functional head needs to license subjects (assign case, satisfy EPP), it must bear [$u\phi$] that gets valued by the subject's [$i\phi$] through upward search. Balti Asp should bear [$u\phi$] and receive valuation from subject DPs. The framework allows for default insertion when upward probing fails (Bjorkman & Zeijlstra, p. 372–374), predicting that Balti should show consistent default ϕ -morphology (e.g., third person singular, cross-linguistically common as default (Bjorkman & Zeijlstra, p. 369–371)).

What we observe instead: Balti shows no ϕ -morphology whatsoever—not defaults, not failed agreement markers, but complete absence of person-number marking on the pivot. The auxiliary *yod* is invariant not because it shows "default third person" but because it encodes only aspect-evidentiality (perfective + egophoric) without any ϕ -specification.

Moreover, Balti permits extensive pro-drop without ϕ -agreement morphology:

(2) *(Pro) aasho thong-sad pa*
 woman see-PST.PFV
 '(Someone) saw the woman'

Upward Agree predicts pro-drop should correlate with successful upward valuation (Bjorkman & Zeijlstra 2019, p. 367–369), yielding visible ϕ -morphology to license null subjects (Zeijlstra 2012, p. 502–503). However, Balti shows pro-drop without any ϕ -morphology, requiring only aspectual-evidential marking plus discourse recoverability. This discourse-based Pro Drop dissociates null-subject licensing from ϕ -agreement entirely, contradicting Upward Agree's prediction that [$u\phi$] valuation is necessary for argument licensing.

The MPH analysis: Balti Asp bears no ϕ -features whatsoever. The pivot values case via [i Case] and satisfies EPP via [u EPP] through obligatory movement of the subject DP to Spec-Asp/EvidP, with no ϕ -probing initiated. Subject DPs retain their [$i\phi$] as freestanding interpretable features accessible to LF interpretation. Pro-drop is licensed through discourse recoverability when context makes referents salient, not through morphological ϕ -agreement. No crash occurs because no ϕ -features require valuation; no defaults appear because there are no ϕ -features to receive default values.

Empirical Challenge 2: Hindi-Urdu [$i\phi$] Pivots

Hindi-Urdu Asp bears [$i\phi$] rather than [$u\phi$], engaging in matching-based Agree rather than valuation:

(3) Imperfective:
(pro) roti khā-tā hai
 bread.F eat-IMPF.M.SG be.3SG
 '(He) eats bread'

The imperfective suffix *-tā* shows masculine singular morphology reflecting Asp[$i\phi$] matching with nominative subject's [$i\phi$]. This licenses morphologically mediated pro-drop.

The challenge for Upward Agree: Under Bjorkman & Zeijlstra (2019, p. 360–361), functional heads bear [$u\phi$] that get valued by DP's [$i\phi$]. Hindi-Urdu reverses this: Asp bears [$i\phi$] (evidenced by rich paradigmatic morphology: *-tā* M.SG, *-tī* F.SG, *-te* M.PL) and matches with DP's [$i\phi$]. There is no [$u\phi$] on Asp requiring upward valuation. Instead, both Asp and DP bear interpretable features that must be compatible and co-indexed through matching.

Upward Agree could respond that DPs bear [$u\phi$] that probe upward to value against Asp[$i\phi$], but this is unmotivated: DPs inherently bear interpretable ϕ -features (person, number, gender) that do not require valuation. The Hindi-Urdu pattern requires

bidirectional compatibility checking between Asp's[i ϕ] and DP's[i ϕ], not unidirectional upward valuation.

Perfective asymmetry compounds the problem:

(4) Perfective:

(*pro*) *roṭī khā-ī*

bread.F.SG eat-PFV.F.SG

'(Someone) ate bread'

Perfective Asp still bears [i ϕ], but ergative case on subjects blocks matching, redirecting Asp to match with the nominative object *roṭī* (yielding feminine singular *khā-ī*). Upward Agree would need to posit that ergative case triggers intervention blocking upward search from continuing to the subject, forcing the DP to search laterally or downward to the object—but this abandons the theory's core claim of uniform upward directionality.

Conclusion: Upward Agree successfully captures phenomena where lower elements appear sensitive to higher specifications (negative concord, sequence-of-tense).

However, it faces challenges in dealing with ϕ -less systems where pivots license arguments without ϕ -mediation.

4.2 Flexible Agree: Preminger 2014

Flexible Agree proposes that unsuccessful ϕ -feature valuation does not crash the derivation (Preminger 2014, p. 6–9). When a probe bearing [u ϕ] fails to find a suitable goal, the derivation continues with unvalued features either deleted at the interfaces or realized as default morphology (Preminger 2014, p. 86–92, p. 126–133). This accounts for cases where agreement appears optional or where defaults surface despite the presence of potential agreement controllers.

The core challenge: Flexible Agree assumes pivots universally bear [u ϕ] and that surface variation reflects valuation success vs. failure. The MPH demonstrates systems where pivots bear no ϕ -feature, with obligatory rich morphology encoding non- ϕ specifications (aspect, evidentiality) and discourse-licensed pro-drop, patterns incompatible with failed valuation.

Empirical Challenge 1: Balti Obligatory Evidentials Without ϕ -Features

Balti aspectual-evidential morphology is obligatory, paradigmatically rich, and entirely ϕ -unavailable:

(5) Balti evidential paradigm (all forms lack person-number distinctions)

a. *go song-set* '(He/she/they) went' (perfective direct)

b. *go-en yod* '(He/she/they) is going' (progressive visible)

c. *go-ain yod* '(I/we) am/are going' (progressive egophoric)

d. *go-chatkhikha song-fin* '(He) must have gone' (inferential)

e. *go-song-fin* '(They say) (he) went' (reportative)

Crucial observation: Egophoric (21c) indicates the speaker's direct experience, not ϕ -agreement. The "1st person" interpretation derives from evidential semantics (speaker-involvement), not from morphological person marking. No form shows person, number, or gender distinctions—the verb stem and evidential markers remain invariant across all ϕ -specifications.

The challenge for Flexible Agree: The framework predicts that when [u ϕ] fails to find goals, two outcomes are possible:

1. Default morphology: Uniform (typically 3SG) marking across contexts (Preminger 2014, p. 88–92)

2. Optional morphology: Probe features surface only when valuation succeeds

Balti shows neither pattern. Evidential morphology is:

- Obligatory: Every finite clause requires evidential specification (*go* 'went' without evidential is ungrammatical)
- Paradigmatically rich: Five distinct evidential categories with aspectual contrasts
- Entirely φ -less: No person-number morphology exists, not even as "defaults"

Under Flexible Agree, obligatory rich morphology should result from successful $[\text{u}\varphi]$ valuation (yielding agreement) or from probe features themselves being spelled out (yielding default 3SG). Balti's obligatory paradigms reflect aspectual-evidential features on the pivot, not φ -valuation success/failure.

Pro-drop compounds the problem: Flexible Agree predicts that morphologically licensed pro-drop correlates with successful φ -valuation (p. 251–258), since default morphology (resulting from failed valuation) typically lacks richness to identify null subjects (p. 133). When pro-drop occurs despite failed valuation, additional licensing mechanisms (discourse salience, topic continuity) are required.

Balti permits extensive pro-drop:

- (6) *(Pro) zan za-en yod*
 food eat-PROG EVID.NPST
 '(He/she) is eating food'

Pro-drop requires evidential + discourse context, rather than φ -agreement.

Flexible Agree cannot straightforwardly derive this: obligatory evidential paradigms without φ -interaction licensing discourse-based pro-drop suggests no φ -probing occurs, not failed valuation with defaults.

Empirical Challenge 2: Expletive Requirements

Flexible Agree must explain why English requires expletives when no thematic subject is available, while Balti permits null subjects freely:

- (7) English vs. Balti weather predicates
 a. English: *(It) is raining. → Expletive obligatory
 b. Balti:
Charfa ongain yoda? → No Expletive required
 Rain come PROG. DIR. Q
 'Is rain coming?'

Under Flexible Agree, English weather verbs lack thematic subjects, Balti employs the weather itself as the subject DP along with a common verb such as *ongain* (coming). The framework must stipulate language-specific conditions determining whether failed valuation requires expletive insertion (English) or permits weather to serve as the subject itself (Balti). This loses the generalization that expletive requirements correlate with probe φ -specifications.

Bimodal Agree derives the contrast from featural architecture: English T bears $[\text{u}\varphi]$: Creates a valuation dependency requiring either a DP goal with $[\text{i}\varphi]$ or expletive insertion. Uninterpretable features must be eliminated before LF (Full Interpretation), forcing overt realization even with non-thematic expletives.

Balti Asp bears $n\varphi$: Eliminates the valuation dependency entirely. EPP is satisfied by obligatory projection of aspectual-evidential morphology hosting the subject; no φ -features require checking. Subject φ -content is recovered from discourse when contextually salient. No expletive is needed because no φ -valuation is demanded.

Why Flexible Agree Cannot Simply Posit φ -unavailable Systems: Flexible Agree could accommodate Balti by allowing pivots to optionally lack $[\text{u}\varphi]$ entirely, eliminating valuation dependencies. However, this concedes the MPH's core claim: pivots vary in φ -specification $[\text{n}\varphi]$, $[\text{u}\varphi]$, $[\text{i}\varphi]$) rather than universally bearing $[\text{u}\varphi]$ with parametric variation in valuation outcomes. Moreover, Flexible Agree's theoretical motivation—

explaining why failed valuation doesn't crash—becomes vacuous if pivots can simply lack features that trigger valuation. The framework's explanatory force depends on universal $[u\phi]$ on pivots attempting valuation, with cross-linguistic variation arising from success/failure patterns rather than from featural specifications themselves.

Conclusion: Flexible Agree successfully handles variation within ϕ -active systems where valuation may succeed or fail. However, it faces challenge in accommodating $[n\phi]$ systems where pivots obligatorily project rich paradigms (evidentials) without any ϕ -interaction, license arguments through case alone, and permit discourse-based pro-drop without morphological ϕ -content. The Balti pattern requires recognizing that some pivots lack ϕ -features entirely, not that they bear $[u\phi]$ with consistently failed valuation.

4.3 Feature Inheritance: Chomsky 2007, 2008; Richards 2007; Miyagawa 2010

Feature Inheritance proposes that ϕ -features originate on phase heads (C, v), and transfer to non-phase heads (T, V) where they trigger agreement (Chomsky 2007, p. 10–12, Chomsky, 2008, p. 138–143). Richards (2007, p. 138–142) extends this to successive-cyclic movement, while Miyagawa (2010, p. 59–66) allows ϕ -features to remain on C in discourse-configurational languages. This framework elegantly derives correlations between phasehood and agreement while maintaining T's central role in finite clauses.

The fundamental incompatibility: Feature Inheritance requires TP projection in all finite clauses, since finiteness correlates with ϕ -feature activity inherited to T (Chomsky 2008, p. 143–147). The MPH demonstrates that finite clauses can lack TP entirely when non-T heads serve as pivots.

Empirical challenge 1: Hindi-Urdu subjunctives (AspP without TP)

(8) *maiñ chāhtā hūñ [ki Abdullah roṭī khā-e]*

I want.M.SG be.1SG that Abdullah bread eat-SBJV.M

'I want that Abdullah eat bread'

The embedded clause is finite (licenses nominative *Abdullah*, forms independent conjuncts, selected by finite *ki*) yet contains only aspectual morphology. The subjunctive *-e* contrasts paradigmatically with perfective *-ā* and imperfective *-tā*, not with tense: temporal interpretation depends entirely on matrix *chāhtā hūñ*, with tense expressed by optional auxiliaries (*thā* past, *gā* future). Aspectual morphology is obligatory; tense is not. This distribution indicates Asp, not T, anchors the clause.

Feature Inheritance must claim *-e* occupies T. However: (i) *-e* patterns aspectually, not temporally; (ii) embedded temporal interpretation⁴ is non-local, derived from the matrix verb rather than independent embedded T; (iii) cross-linguistically, subjunctive is aspectual in Hindi-Urdu but fused with T in Italian (*vada* SBJV.PRS vs. *andassi* SBJV.IMPF), indicating subjunctive does not universally occupy T.

Empirical challenge 2: Italian imperatives (MoodP without TP)

(9) *Vieni e siediti!*

come.IMP.2SG and sit.IMP.2SG=REFL

'Come and sit down!'

Italian imperatives show mood morphology without tense distinctions (**X venne* past.IMP, **X verrai* fut.IMP) yet function as independent finite clauses. Crucially, clitic placement is mood-conditioned:

⁴ However, it must be noted that there can be clauses in which neither the matrix nor the embedded clauses contain T, i.e., *maiñ-ne chāha [ki Abdullah roṭī khā-e]*. Neither of the clause contain a temporal marker but both contain aspectual inflections. This further establishes the subservient nature of TP and superiority of Asp in Hindi-Urdu.

- (10) a. *Lo vedo.* (indicative: proclitic)
 it.CL see.IND.1SG 'I see it'
 b. *Vedilo!* (imperative: enclitic)
 see.IMP.2SG=it.CL
 'See it!'

If T universally hosted inherited ϕ -features as proposed by Chomsky (2008, p. 143–147), clitic placement should be uniform. The mood-conditioned alternation demonstrates that Mood, not T, determines structural organization.

Feature Inheritance would require positing imperative-as-T, but this fails to explain: (i) absence of tense paradigms within imperatives; (ii) mood-conditioned syntax extending across multiple phenomena (subject position, negative imperatives, complementizer selection); (iii) why clitic position tracks mood rather than tense (see Appendix E.4.1)

Miyagawa (2010) does not resolve the issue: While Miyagawa allows ϕ -features to remain on C in discourse-configurational languages, his framework still assumes ϕ -features correlate universally with C or T (2010, p. 32–41). The MPH shows ϕ -features correlate with Asp in Hindi-Urdu (gender-number agreement tracks aspectual morphology: *khā-yā* M.SG vs. *khā-ī* F.SG), with Mood in Italian (subjunctive agreement tracks mood paradigms), and are absent entirely in Balti (aspectual-evidential pivots without person-number marking). Accommodating these patterns requires positing C-to-Asp or C-to-Mood inheritance, conceding the MPH's core claim that different functional heads anchor clause structure cross-linguistically.

Feature Inheritance maintains that T universally hosts finiteness-related features (whether inherited from C or residing there directly), predicting TP in all finite clauses. The MPH demonstrates that AspP, MoodP, or AgrP can satisfy all licensing requirements without TP projection. Even flexible formulations (Richards 2007; Miyagawa 2010) face challenges in accommodating this variation without substantially weakening the assumption of T-centrality underlying Feature Inheritance.

4.4 Multiple Agree (Hiraiwa: 2001, 2005)

Multiple Agree proposes that a single probe can establish agreement relations with multiple goals simultaneously, capturing polypersonal agreement patterns where morphology reflects both subject and object ϕ -features (Hiraiwa 2001, p. 68–70; 2005, 99–108; Nevins 2011).

The core challenge: Hindi-Urdu demonstrates that apparent "object agreement" involves case-conditioned selective matching with one accessible goal, not simultaneous Multiple Agree with all arguments.

Hindi-Urdu: case-conditioned selective matching

Hindi-Urdu Asp bears [$i\phi$] in both imperfective and perfective contexts. However, case assignment creates crucial asymmetries in which arguments can match with Asp:

Imperfective: Nominative subjects successfully match with Asp[$i\phi$]:

- (11) *Abdullah roṭī khā-tā hai*
 Abdullah bread.F.SG eat-IMP.F.M.SG be.PRS
 'Abdullah eats bread' → -*tā* matches masculine singular *Abdullah*

Perfective: Ergative case blocks ϕ -matching between subject[$i\phi$] and Asp[$i\phi$] (Bobaljik 1993; Laka 1993; Woolford 1997). Asp redirects matching to the accessible object:

- (12) *Abdullah-ne roṭī khā-ī*
 Abdullah-ERG bread.F.SG eat-PFV.F.SG
 'Abdullah ate bread' → -*ī* matches feminine *roṭī*, not ergative *Abdullah* M.SG

Critical pattern 1: Default with inaccessible objects

When objects are non-specific or absent, default masculine appears despite the ergative subject bearing [iφ]:

(13) *Abdullah-ne (kuch) khā-yā*

Abdullah-ERG something eat-PFV.M.SG

'Abdullah ate (something)' → default masculine (failed matching)

Multiple Agree predicts that if Asp probes both subject and object simultaneously, morphology should reflect attempted multiple probing even when one goal is inaccessible (Hiraiwa 2005, p. 118–135). Instead, we observe uniform default morphology indicating complete matching failure, not simultaneous partial agreement.

Critical pattern 2: Only one argument controls morphology

When objects are accessible, morphology reflects only object φ-features with no morphological trace of ergative subjects:

(14) *Abdullah-ne rotīyān khā-ī*

Abdullah-ERG breads.F.PL eat-PFV.F.PL

'Abdullah ate breads' → only object controls morphology

Multiple Agree in polypersonal languages (Basque, Georgian) yields morphological reflexes of both subject and object φ-features simultaneously. Hindi-Urdu shows exclusive matching with one accessible argument, indicating case-conditioned intervention rather than Multiple Agree.

The MPH analysis: Hindi-Urdu Asp[iφ] engages in sequential probing where case determines accessibility. Ergative case blocks φ-matching with subjects despite both bearing [iφ]; Asp successfully matches with objects when accessible; failed matching yields default morphology (Preminger 2014). This is selective matching conditioned by case, not Multiple Agree with all arguments.

Warlpiri: Templatic Agreement vs. Free Word Order

Warlpiri auxiliaries show fixed subject-object agreement morphology (*ka-rna-ngku* PRES-1SG.SUBJ-2SG.OBJ) occupying rigid second position, while DPs exhibit completely free word order:

(15) a. *Ngarrka-ngku ka-rna-ngku nya-nyi kurdu-ku*

man-ERG PRES-1SG.S-2SG.O see-NPST child-DAT

b. *Kurdu-ku ngarrka-ngku ka-rna-ngku nya-nyi*

[Same meaning, free DP order]

Multiple Agree predicts simultaneous probing should correlate with DP structural positions. Instead, morphological templatic order (subject > object) remains constant while DP order varies freely, and the auxiliary position (second position) is independent of both. This suggests the auxiliary establishes agreement in situ within vP (yielding templatic morphology reflecting base hierarchy), then undergoes head-movement to second position (Legate 2002), satisfying [uEPP] through pivot movement rather than attracting phrases. Multiple Agree cannot dissociate agreement configuration from surface positions.

Conclusion: Multiple Agree successfully captures polypersonal agreement in configurational languages where probes simultaneously access all structurally available goals. However, it faces challenges in dealing with case-conditioned selective matching where ergative case blocks φ-matching despite both probe and goal bearing [iφ] (Hindi-Urdu), or head-movement EPP satisfaction dissociating agreement from word order (Warlpiri). The MPH derives these patterns through Bimodal Agree with case-based intervention and parametric EPP mechanisms.

4.5 Theoretical significance of the φ-parameterization

The proliferation of auxiliary mechanisms in recent minimalist work—Feature Inheritance (Chomsky 2008), Feature Sharing (Pesetsky & Torrego 2007), Upward Agree (Bjorkman & Zeijlstra 2019), and Failed Agree (Preminger 2014)—reveals an unresolved gap in the theory of φ -agreement. Each proposal introduces additional machinery to accommodate empirical configurations that a single undifferentiated notion of φ cannot capture. In effect, these mechanisms acknowledge that φ -features are not uniform objects: they vary in their availability on functional heads and in whether their values contribute to interpretation at LF.

The present analysis makes this implicit distinction explicit by formalizing it within a two-tiered φ -parametric architecture, validated both statistically (§3.5.1–3.5.4) and against independent corpus data external to the theory (§3.5.5). Once φ -availability and φ -interpretability are parameterized as distinct choices, the empirical domains that previously required inheritance, sharing, or repair fall out from general principles. This reframing supports viewing MPH as a promising integrative framework: it identifies the latent structural dimension that earlier frameworks were converging on from different directions. The dual-validated two-tiered φ -parameters — corroborated by both internal statistical structure and external corpus data — supply the missing generalization, restoring explanatory continuity across the Agree-based literature.

5. Counterarguments and Theoretical Implications

5.1 Addressing Standard Objections

The framework proposed here anticipates several theoretical objections that may be raised against pivot-based analyses more generally. This section demonstrates that the Hierarchical Parametric Organization withstands these objections while offering empirical advantages over competing approaches.

The circularity objection raised by Legate (2008) and Baker (2015) holds that identifying the functional head hosting the EPP bundle risks defining the pivot as "the head that licenses the subject," making EPP-related behavior trivially follow. The present framework explicitly breaks this circle through its Biphasic Identification Protocol: the pivot is identified syntactically first, through independent clause-boundary and movement diagnostics, and only then corroborated morphologically via Bybee's (1985) criteria. Only heads passing at least two diagnostics from each family qualify for morphological testing, making pivot selection replicable and falsifiable across analysts and datasets.

The universal tense hypothesis, common in Minimalist literature (Chomsky 2001, 2007), assumes the EPP is universally hosted by T. Cross-linguistic evidence demonstrates otherwise: Aspect serves as pivot in Indo-Aryan languages, Agreement in Bantu, Mood in Romance, fused Aspect-Evidential systems in Balti. The EPP Hosting Maxim derives this distribution without stipulating universal T — a refinement of the feature-anchoring principle, not an abandonment of Minimalist architecture: the EPP is universally required but not universally located.

Feature Inheritance (Chomsky 2007, 2008) faces difficulty explaining why φ -features are entirely absent on Balti Asp/Evid, or why φ -interactions correlate with morphological richness across unrelated families. Under the two-level architecture, φ -availability is a lexical property of the pivot itself, not a result of top-down inheritance. Feature Inheritance predicts uniform φ -behavior across finite clauses; the attested cross-linguistic split in φ -availability confirms the parametric account instead. Preminger's (2014) Flexible Agree predicts valuation failure should not crash derivations, but does not predict the systematic coupling of case, EPP, and φ -availability found here: in Balti, φ -valuation never initiates yet EPP and case are obligatorily

realized; in English, φ -valuation is necessary for EPP satisfaction. This polarity requires Bimodal Agree, which Flexible Agree lacks a mechanism to derive.

The unreliable morphology objection (Harley & Noyer 1999) misunderstands morphology's role here: Bybee's criteria corroborate syntactic findings independently obtained through structural diagnostics, rather than inferring structure from form. Where morphology and syntax diverge — Balti evidentials lacking φ -agreement yet enforcing EPP; German, Icelandic, Faroese, and Finnish retaining rich agreement despite uninterpretable φ (§3.5.2–3.5.3) — the analysis treats this as crucial evidence, not noise to explain away. Morphology plays a diagnostic, not deterministic, role.

The parameter proliferation objection might argue that positing two parameters increases complexity relative to single-parameter accounts of pro-drop or agreement. The two-level architecture instead reduces total parameterization: traditional accounts require independent parameters for pro-drop availability, agreement richness, expletive requirements, case systems, and EPP satisfaction. MPH derives all five from two binary choices, capturing their tight cross-linguistic clustering (PC1 = 96.60% variance explained, §3.5.2) and independently confirmed via corpus data external to the theory (§3.5.5). The resulting system obeys Minimalist economy: one locus (the pivot), one feature class (φ -availability with binary interpretability).

Mixed systems like Spanish and Turkish, showing both φ -agreement and discourse-based pro-drop, are predicted patterns, not counterexamples: hybrid Agree combines interpretable φ -features on pivots with recoverable goal φ -features in discourse. Non-finite clauses hosting subjects without overt φ -agreement (English infinitives) represent reduced projections lacking fully specified pivots, with [iCase] remaining on the matrix head.

One methodological concern warrants explicit discussion: the freezing diagnostic proved systematically weaker than other movement tests across all languages examined (§2). English, Swahili, Italian, and Balti all showed modest or inconclusive freezing effects ($\kappa=0.69$ — the lowest of the six diagnostics; Appendix C). This reflects cross-linguistic variation in extraction strategies and possessor-extraction confounds, not weakness in pivot identification itself — no language's pivot status depends solely on freezing effects, since the nomination rule requires ≥ 2 movement diagnostics. The remaining five diagnostics — coordination ($\kappa=0.89$), complementizer selection ($\kappa=0.84$), ellipsis ($\kappa=0.79$), extraction ($\kappa=0.77$), and subject position ($\kappa=0.73$) — show substantially higher reliability (overall $\kappa=0.79$, 95% CI [0.76, 0.82]; Appendix C), ensuring robust identification; freezing results receive appropriate weight as one diagnostic among several rather than a load-bearing criterion.

None of the standard counterarguments successfully undermines the proposal, as each fails to predict the cross-linguistic correlation between φ -availability, case licensing, and pro-drop that Bimodal Agree captures directly.

5.2 The Objectivity of the Coding Schema and the PCA Results

A legitimate methodological concern is whether the Principal Component Analysis is circular, given that its input variables derive from MPH itself. Three considerations demonstrate that the analysis tests the theory rather than restating it.

First, the variables are operationalized through objective, replicable diagnostics, each defined by language-independent empirical thresholds rather than theoretical stipulation: Rich_Agreement via Bybee's (1985) criterion (≥ 3 contrastive forms); Morph_ProDrop via story-initial elicitation yielding $\geq 80\%$ speaker acceptance of null subjects; No_Expletive via context-neutral weather-verb constructions. These diagnostics were applied consistently across all languages. Unlike the Phase 1 syntactic

nomination diagnostics, which involve graded grammaticality judgments and were accordingly validated through independent double-coding (15% subset, $n=1,200$ items; overall $\kappa=0.79$, 95% CI [0.76, 0.82], Appendix C), the four PCA-input variables are constructed as fixed quantitative thresholds rather than judgment calls, and their objectivity rests on the reproducibility of the threshold procedure itself: any researcher applying Bybee's ≥ 3 -contrastive-form criterion, the $\geq 80\%$ -acceptance elicitation protocol, or the weather-verb expletive test to the same language should reach the same classification without needing to adjudicate disagreement between coders. This division — inter-rater reliability for the graded Phase 1 diagnostics, procedural reproducibility for the categorical Phase 2 variables — reflects the different epistemic status of the two diagnostic types, not an inconsistency in methodological rigor.

Second, the variables are independently testable constructs. A researcher can apply the diagnostic battery from Appendix D to a new language and generate a coding profile without commitment to MPH or knowledge of its predictions. That this has been done for seventy-five outgroup languages demonstrates the framework's capacity for blind, objective application.

Third — and now decisively, on two independent grounds rather than one — the analysis is corroborated by structure it did not build in.

The internal statistical evidence must be stated carefully. PC1 explains 96.60% of variance across the four hand-coded diagnostics (§3.5.2), and the polychoric correlation matrix (§3.5.3) reaches its estimation boundary (all pairwise coefficients ≈ 1.000) — reflecting the genuine empirical fact that, once corrected, the sample contains essentially no counter-examples on three of the four diagnostics. A sceptical reader could reasonably ask whether concentration this high itself is evidence of circularity. It is not, for a reason that does not depend on PC1's magnitude: **the diagnostics were tested against an evidence source that had no part in generating them.**

Phi_Availability values derived mechanically from raw Universal Dependencies treebank morphology — a corpus built by independent annotators with no relationship to MPH — recover the hand-coded classifications at 95–97.5% accuracy (§3.5.5), using a threshold fixed before the test was run. This is the decisive non-circularity evidence: not an internal property of the PCA (which, on a highly deterministic four-variable system, will always show high PC1 regardless of whether the underlying construct is real), but an external test the theory could have failed and did not.

The internal analysis retains real evidential value, but for a different reason than originally claimed: **PC2, though small (3.40%), is not noise — it isolates exactly one theoretically motivated distinction.** Rich_Agreement contributes 76.83% of PC2's variance, separating Type 2a (impoverished morphology: English, Dutch, French) from Type 2b (retained rich morphology despite uninterpretable status: German, Icelandic, Faroese, Finnish) — precisely the dissociation between agreement richness and interpretability that is this study's most theoretically interesting empirical finding (§3.2.2, §3.5.2). A circular construction, where the four variables were mere restatements of one theoretical claim, would not produce this structured, independently-interpretable residual; it would produce homogeneous noise.

The theory's falsifiable claim must also be stated precisely. As established in §1.9.2, the theory's falsifiable claim excludes Rich_Agreement from its inviolable core: the theory is falsified by any language **where Phi_Availability, Morph_ProDrop, or No_Expletive diverge from one another** (this deterministic trio holds at $r=1.000$ across all 82 sample languages, with no exception). Rich_Agreement is explicitly *not* part of this inviolable core — a language showing Rich_Agreement=1 alongside No_Expletive=0.2 is not a

counterexample but an instance of the predicted Type 2b pattern, independently confirmed in four languages via both internal coding and external UD corpus data (§3.5.5). Stating the falsifiable claim this precisely is a strength, not a hedge: it specifies in advance exactly which combinations are forbidden and which are an expected, independently-motivated exception, and both halves of that specification are borne out empirically.

The decisive test of non-circularity, restated: could a researcher blind to MPH generate the same coding, and would an entirely independent data source agree with it? Both are now answered affirmatively with evidence: the diagnostic battery is fully specified and quantitative (Appendix E), application to seventy-five outgroup languages succeeded without foreknowledge of predictions, and the resulting codings were independently corroborated by Universal Dependencies treebank data the researcher did not construct (§3.5.5). This is a stronger standard of non-circularity than internal statistical structure alone can provide, and the study meets it.

5.3 Typological Predictions and Empirical Scope

The two-level parametric architecture generates specific testable predictions about possible and impossible language types, creating falsifiable boundaries on cross-linguistic variation.

Predicted impossibilities:

- Languages should not show morphologically-licensed pro-drop while requiring obligatory expletive insertion, or vice versa, since these are diagnostics of interpretable and uninterpretable ϕ respectively and cannot co-occur on the same pivot (§3.5.3, §3.8). **Rich person-number agreement morphology, by contrast, is explicitly not predicted to track this distinction perfectly:** German, Icelandic, Faroese, and Finnish show rich agreement alongside expletive-obligatoriness and blocked pro-drop, a stable, independently-confirmed pattern (Type 2b; §3.2.2, §3.5.5) reflecting agreement's capacity to survive diachronically after a pivot's ϕ -features have already become uninterpretable (§5.5) — not a rare or transitional exception, but a predictable consequence of the theory's own claim that Rich_Agreement is dissociable from the other three diagnostics.
- Languages should not lack person-number/gender morphology entirely while showing morphologically-mediated pivot-based pro-drop, since morphological licensing requires ϕ -information recoverable from the pivot.
- Languages should not show discourse-insensitive pro-drop without agreement morphology, since only pivot-mediated pro-drop through interpretable ϕ -features permits pro-drop independent of discourse context.

Predicted property clustering:

All ϕ -unavailable systems should uniformly lack person-number/gender paradigms on pivots, show aspectual or evidential morphology as clause-defining, exhibit aspect-conditioned (not ϕ -conditioned) case alternations, permit discourse-mediated pro-drop with person-hierarchy effects, and show topic-structure sensitivity in null-subject licensing.

All ϕ -available systems with interpretable features should show rich agreement paradigms, permit pro-drop without discourse restrictions, lack expletive requirements, and show ϕ -interacting case systems.

All ϕ -available systems with uninterpretable features should require overt subjects or expletives, block pro-drop, and tie case licensing to ϕ -valuation — **but agreement richness within this type is not uniform: Type 2a (English, Dutch, French) shows**

impoverished agreement; Type 2b (German, Icelandic, Faroese, Finnish) retains rich agreement despite uninterpretable status. This split is itself a confirmed prediction, not a complication: it follows directly from Rich_Agreement's independent diachronic behavior (§3.5.6, §5.5).

Extensibility to other systems: Because pivothood is empirically defined through replicable diagnostics, the model extends naturally to other alignment systems — Philippine-type voice systems, Agent-Focus (A'-driven) systems, Quechuan switch-reference. Each requires the same Biphasic protocol: identifying which functional head passes clause-boundary and movement diagnostics, then corroborating through morphological prominence. Systematic investigation of understudied families is predicted to reveal additional pivot types, potentially including voice-based pivots, evidential-only pivots without aspectual specification, or mood-aspect fusions distinct from the Italian pattern.

Bounded typological diversity: The framework predicts bounded, not unlimited, typological diversity: three surface types (0.1, 0.2, 1.0), with Type 2 internally structured into 2a/2b by agreement richness rather than constituting a fourth macro-type. Mixed or transitional systems like Spanish and Turkish represent parameter interaction or diachronic drift in φ -availability, not additional types. Surface changes in agreement richness are predicted to occur more frequently than changes in fundamental clause-anchoring mechanism, since shifts in φ -interpretability within φ -available systems require less radical reanalysis than shifts between φ -available and φ -less systems — a prediction for which Type 2b now constitutes direct, attested confirmation (§5.5), not merely a theoretical possibility.

5.4 Acquisition and Processing Implications

Learnability follows directly from surface-visible morphological cues, supporting minimalist acquisition models emphasizing distributional learning from primary linguistic data.

Children acquiring φ -available systems with interpretable features should master agreement paradigms early, produce appropriate pro-drop from the earliest multiword stage, and show no extended period of expletive-omission errors, since expletives are unnecessary. Swahili acquisition studies (Deen 2003, 2006; Demuth 2003) confirm this: subject prefixes appear by eighteen months, with near-errorless agreement mastery by age two and null subjects produced appropriately from the start.

Children acquiring φ -available systems with uninterpretable features should produce null-subject errors during an Optional Infinitive stage before recognizing overt subjects are required and show extended expletive-omission errors before mastering that expletives satisfy EPP requirements. English acquisition confirms this: null subjects during the Optional Infinitive stage around ages two to three, late expletive-insertion acquisition around ages three to four (Pérez-Leroux & Wexler 1999), with extended omission errors before convergence on obligatory overt subjects by ages four to five.

Critically, this delay is predicted to track interpretability rather than paradigm richness (§3.8): German — despite rich adult agreement — shows a comparable Optional Infinitive stage to English, distinguishing this account from one where delay simply reflects paradigm poverty.

Children acquiring φ -less systems are predicted to master aspectual and evidential morphology early, since it is the clause-defining system, show discourse-appropriate pro-drop from early stages, and potentially overgeneralize pro-drop to discourse-inappropriate contexts before fully acquiring pragmatic licensing conditions. Acquisition

data for Balti and related Tibetan languages remain limited; these predictions are testable through future fieldwork and provide clear falsification criteria.

Surface visibility and early parameter setting supports Wexler's (1998) Very Early Parameter Setting hypothesis: rich paradigms provide direct positive evidence for interpretable φ -features; sparse paradigms combined with obligatory expletives provide direct positive evidence for uninterpretable φ -features **in Type 2a systems specifically** — for Type 2b, the relevant positive evidence is expletive-obligatoriness and blocked pro-drop *despite* rich paradigms, since paradigm richness alone would misleadingly suggest interpretability. Absence of φ -morphology combined with discourse-conditioned null subjects provides direct positive evidence for φ -less pivots. Convergence is predicted by ages two to three across all systems.

5.5 Diachronic Trajectories and Language Change

The hierarchical parametric structure predicts asymmetric patterns in language change. Shifts in φ -interpretability within φ -available systems may occur through paradigm erosion or enrichment without fundamentally restructuring clause architecture. **This is not merely a theoretical possibility: it is directly attested in the Type 2b cluster.**

German, Icelandic, Faroese, and Finnish retain rich, productive agreement morphology inherited from earlier interpretable stages, while their pivots have independently become uninterpretable, as confirmed by expletive-obligatoriness and blocked pro-drop (§3.2.2, §3.5.5) — morphology and interpretability have decoupled mid-transition, with agreement retention outlasting interpretability loss. This is the predicted asymmetry made concrete: agreement richness is conservative, changing more slowly than the syntactic property it once reliably signalled. Conversely, analogical extension of paradigms could shift systems from uninterpretable toward interpretable φ -features, potentially enabling pro-drop acquisition as morphological licensing becomes possible, though no case in the current sample is analyzed as undergoing this direction of change. Shifts between φ -available and φ -less systems require more radical reanalysis, potentially driven by reinterpretation of aspectual or evidential morphology. When aspectual markers become obligatory through grammaticalization, learners might reanalyse them as clause-defining rather than optional tense modifiers, potentially triggering a shift from φ -based to aspect-based clause anchoring if agreement paradigms simultaneously erode. Conversely, grammaticalizing person-number markers from pronouns or auxiliaries could trigger the reverse shift, from aspect-based to φ -based anchoring.

The Hindi-Urdu case: the split between perfective (ergative case, object agreement) and imperfective (nominative case, subject agreement) contexts could reflect either stable aspectual conditioning (as in Georgian) or an intermediate diachronic stage. Historical evidence from Indo-Aryan (Butt 2001; Verbeke & De Cuypere 2009) shows ergative patterns arising through reanalysis of perfective participles, suggesting Hindi-Urdu represents a stable split-ergative system rather than a transitional state, predicting continued coexistence of both patterns.

The Italian case: mood and tense have fused extensively while mood retains structural dominance, likely reflecting gradual phonological convergence between originally distinct paradigms. The clitic-placement diagnostic's mood-sensitivity despite this fusion suggests syntactic structure can remain stable even as morphological exponence becomes less transparent — predicting Italian-type systems may be transitional, resolving either toward renewed morphological differentiation or complete fusion.

5.6 Connections to Broader Theoretical Debates

The present framework engages with ongoing debates about whether Minimalism should maintain universal functional hierarchies or embrace more radical cross-linguistic variation.

Cartographic approaches (Rizzi 1997; Cinque 1999; Belletti 2004) propose elaborate universal hierarchies with fixed ordering; others argue for flexible, language-specific functional structures. This study occupies middle ground: certain functional roles are universal — something must host the EPP bundle and license case — while the identity of that head varies parametrically. This avoids empirical problems facing rigid universal hierarchies confronted with missing or reordered projections, without abandoning the Minimalist commitment to constrained variation through parameters on universal features.

Morphology and syntax: the claim that morphological prominence correlates with syntactic pivothood bears on whether morphological exponence reflects underlying syntactic structure or can dissociate from it more freely. The framework takes a middle position: morphology corroborates syntactic structure identified through independent diagnostics rather than determining it directly. Where syntax and morphology diverge — Balti's aspectual pivots lacking φ -agreement entirely;

German/Icelandic/Faroese/Finnish's rich agreement outlasting interpretability (§5.5) — this reveals crucial information about feature specification and diachronic timing rather than constituting problematic exceptions.

Bimodal-Agree and syntactic operations: demonstrating that case valuation can proceed independently of any φ -interaction, when pivots lack φ -features entirely, suggests featural content on functional heads varies more radically cross-linguistically than frameworks assuming universal φ -bearing T predict — implying syntactic operations themselves may be partially parameterized rather than purely universal with variation confined to lexical feature specification.

Syntax-pragmatics interface: the distinction between pivot-mediated and discourse-mediated pro-drop bears on the division of labour between narrow syntax and the interfaces. If some languages license pro-drop through morphological agreement and others through discourse structure, LF-interpretation and PF-realization can compensate for each other in satisfying recoverability requirements — predicting that languages with interpretable φ on their pivots show discourse-independent pro-drop, while φ -less-pivot languages show discourse sensitivity, a testable hypothesis about when syntax versus pragmatics determines grammaticality.

Conclusion

The findings converge on a principle with far-reaching implications for syntactic theory: the EPP is not a fixed projectional property of T but a derivative property of the morphosyntactic pivot, identified through convergent syntactic and morphological diagnostics. Once the pivot is independently diagnosed, its φ -availability determines the mode of Agree and consequently the distribution of subjects, case, agreement, expletives, and pro-drop across languages. This unified perspective replaces language-specific stipulations with a two-level parametric architecture linking micro-variation in agreement morphology to macro-variation in clause structure.

The analysis restores symmetry between syntax and morphology by demonstrating that structural centrality and morphological prominence converge on the same functional head. The Biphasic Identification Protocol provides a replicable pathway from surface diagnostics to theoretical generalization, significantly reducing the risk of circularity that has plagued comparative syntax. The EPP, long assumed universal in position,

emerges as universal in requirement but flexible in locus, anchored wherever grammar encodes structural centrality through convergent syntactic and morphological evidence. This claim is now supported by two independent lines of evidence rather than one. The Principal Component Analysis provides robust, quantitative support internal to the coded data: the dominant dimension of variation (PC1 = 96.60%) cleanly separates languages by a single latent parameter — a language's φ -state, whose interpretable- φ setting ([i φ]) is one of its four convergent reflexes. Independently, corpus-derived φ -availability values, computed mechanically from raw Universal Dependencies treebank morphology with no reference to MPH, recover the hand-coded classifications at 95–97.5% accuracy (§3.5.5). The theory's central claim is thus corroborated both by structure internal to its own data and by an external test it could have failed.

Critically, the strongest confirmation of the central theoretical claim comes not from proximity on PC1 but from the behavioural pattern itself (§1.6.3, P3–P4): Balti, a [n φ] pivot entirely lacking φ -features, and the [i φ] languages, whose pivots bear fully interpretable φ -features, converge on identical surface behavior — both permit null subjects and disallow expletives — despite sitting at opposite ends of the φ -Availability parameter. English, a [u φ] pivot, is the outlier: possessing a φ -probe that neither the φ -less Balti nor the φ -interpretable [i φ] languages need in the same way, English alone blocks null subjects and requires expletives. This U-shaped pattern is not predicted by morphological richness — Balti's extensive aspectual-evidential paradigm is more elaborate than English's impoverished agreement, yet English is the behavioural outlier, not Balti — nor by φ -Availability alone, which would place [u φ] English between [n φ] Balti and the [i φ] languages rather than apart from both. It is exactly the signature predicted by φ -Interpretability specifically: the pivotal syntactic distinction is not between systems with 'more' or 'less' morphological structure, but between systems where the pivot's φ -features are semantically active at LF and those where they are not — including the null case where no φ -features are present to begin with.

The secondary dimension (PC2 = 3.40%) then reveals finer structure that PC1 alone does not capture. Rich_Agreement dominates PC2's loading, and its effect is concentrated on isolating Type 2b (German, Icelandic, Faroese, Finnish) — languages whose retained rich agreement sets them apart from every other group, including Type 3, despite Type 2b sharing Type 3's morphological richness and sharing Type 1/Type 2a's uninterpretable or absent φ -status on the primary dimension. PC2 is thus best understood not as a smooth three-way subdivision of the "0" cluster, but as the residual signature of exactly the agreement-richness/interpretability dissociation this study identifies as its central empirical discovery. This geometric description is confirmed directly by the biplot for the corrected 82-language sample (Figure 8): Rich_Agreement contributes 76.83% to PC2, versus 7.72% for each of the other three variables (Table 7), and Type 2b sits well apart from every other group on PC2, exactly as predicted.

This dual-evidence structure also addresses a standard objection to typological PCA studies before it is raised: that apparent clustering reflects genealogical or areal bias rather than a genuine cross-linguistic parameter. A supplementary analysis splitting the sample by Indo-European membership (N=41 each) shows the primary dimension holds independently in both subsamples (Indo-European PC1=92.05%; non-Indo-European PC1=98.51%) — if anything, marginally *stronger* outside the genealogically dominant family. The pattern is not an artifact of the sample's Indo-European weighting. The framework achieves Minimalist economy by deriving multiple phenomena from a single parametric axis. Case systems, agreement patterns, subject licensing, expletives, and pro-drop all follow from φ -availability/interpretability on the pivot rather than

requiring separate parameter- or construction-specific stipulations. The bounded typology generated by the two-level architecture predicts possible and impossible language types while accommodating documented diversity across language families. The surface visibility of morphological cues ensures learnability, while the hierarchical parametric structure predicts asymmetric patterns in diachronic change — patterns for which Type 2b now constitutes direct attested confirmation, not merely a theoretical possibility (§5.5). This dual-validated model — statistically robust and independently corroborated by external corpus data — offers a falsifiable model of clause structure in the classic Popperian sense (§1.9.2): its central predictions could have been contradicted by either line of the evidence and were not.

Beyond the empirical findings about pivot variation, this work contributes to parametric theory by demonstrating that a parameter can exhibit logical dependency, where one parameter's definition is conditional on another's value. The φ -Availability/ φ -Interpretability system generates three surface types (not $2^2 = 4$) through this dependency, with the impossible fourth type ruled out by logical structure rather than by stipulation.

Future research should extend the diagnostic protocols to understudied language families, test predictions about acquisition trajectories in φ -less systems and examine whether additional pivot types exist in languages with different morphological profiles. An attempt was made to extend the corpus-based validation of §3.5.5 to Rich_Agreement, Morph_ProDrop, and No_Expletive using UD's available proxies (agreement-feature richness, null-subject rate, and expl-relation frequency respectively), but this did not succeed: unlike verbal morphology, which UD's FEATS annotation records in a comparatively direct and theory-neutral way, pro-drop and expletive status are structural properties established through syntactic diagnostics (coordination, ellipsis, movement) that a raw count of surface dependency relations cannot reliably approximate — the resulting proxies conflate genuine morphological licensing with topic-drop, imperatives, and treebank-specific annotation conventions, and their inclusion in a multi-factor model degraded rather than improved classification accuracy under honest cross-validation. This suggests corpus-based validation of the full four-variable bundle may require treebanks with richer syntactic annotation than UD's dependency scheme currently provides, rather than being achievable through better feature engineering on the existing resource. Corpus studies tracking diachronic shifts in φ -availability would further illuminate the mechanisms of parametric change directly attested in Type 2b. The primary criterion for evaluating the theory should therefore be the reproducibility of its classifications across independently investigated languages rather than agreement with any individual descriptive tradition. Ultimately, the long-term value of the Morphosyntactic Pivot Theory will be determined by the cumulative results of independent researchers applying the documented protocol across the widest possible range of the world's languages.

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